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**McConnell et al.**

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(54) **WEDGE-LOCK SHEATH RETENTION MECHANISM**

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**A61M 25/00** (2006.01)  
**A61M 25/09** (2006.01)

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(Continued)

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*Primary Examiner* — Kathleen S Holwerda

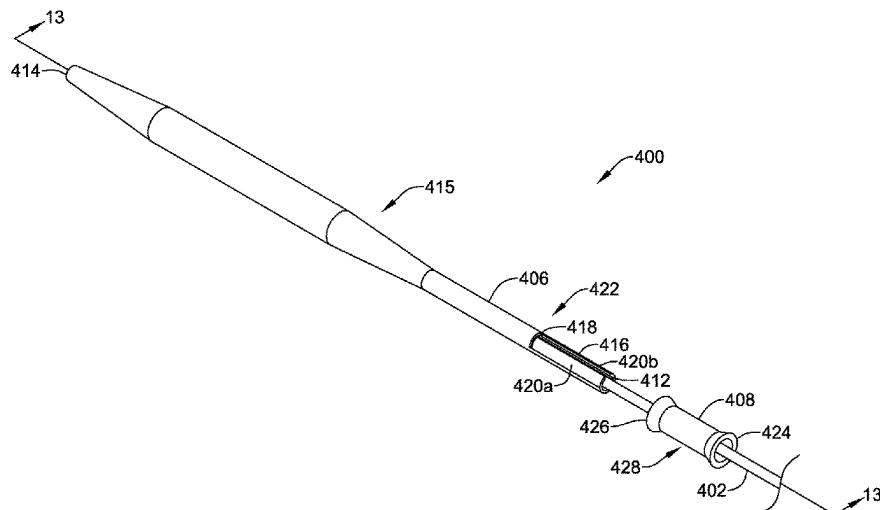
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(57) **ABSTRACT**

A medical device system may include a sheath, a pusher wire, and a locking element. The pusher wire may be slidably disposed within a lumen of the sheath. The locking element may have a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end. The locking element may have a first inner diameter adjacent to the distal end and a second inner diameter adjacent to the proximal end, the second inner diameter smaller than the first outer diameter. The distal end region of the locking element may be configured to freely slide over the sheath and when a proximal end region of the locking element is disposed over the sheath, the locking element may be configured to depress the sheath radially inwards.

**18 Claims, 30 Drawing Sheets**



(58) **Field of Classification Search**

CPC ..... A61B 17/12109–12118; A61B  
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2017/22049; A61B 17/12022; A61M  
2025/09125; A61M 2025/09116; A61M  
25/0102; A61M 25/09  
See application file for complete search history.

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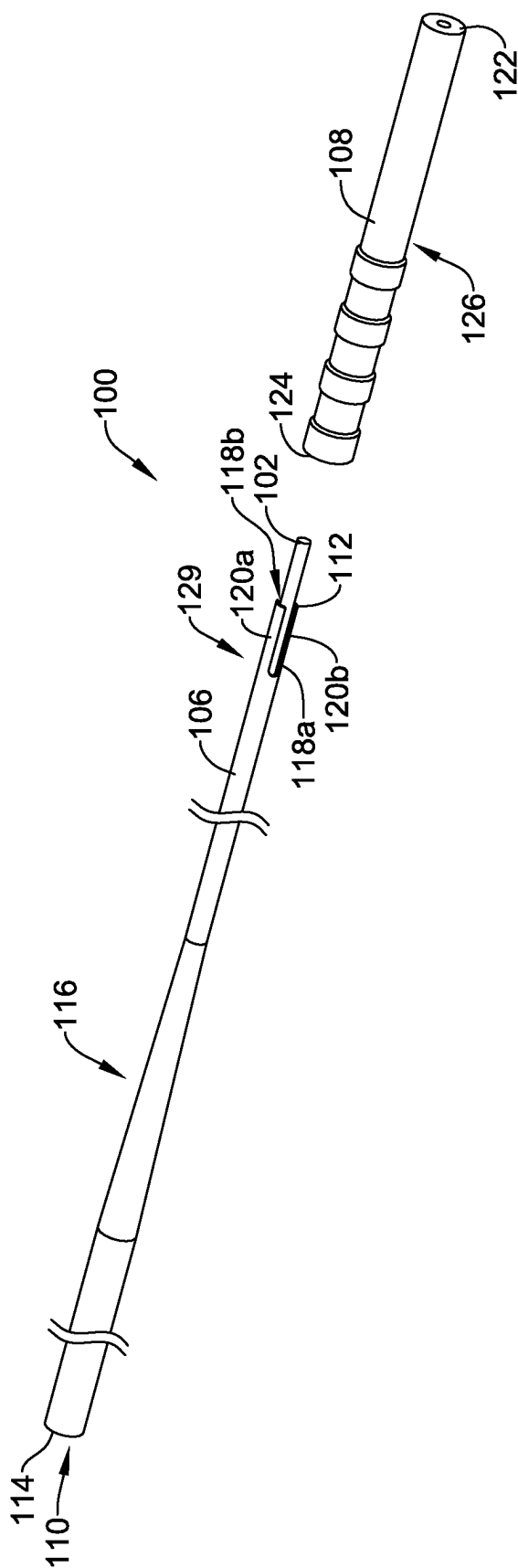
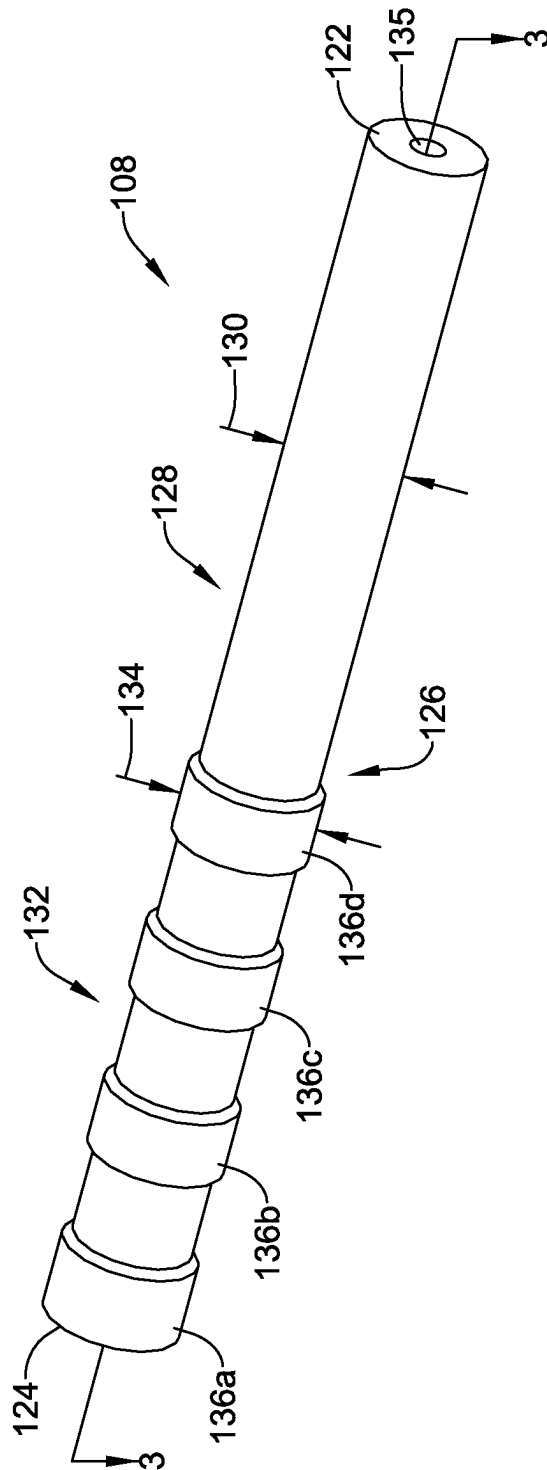


FIG. 1



**FIG. 2**

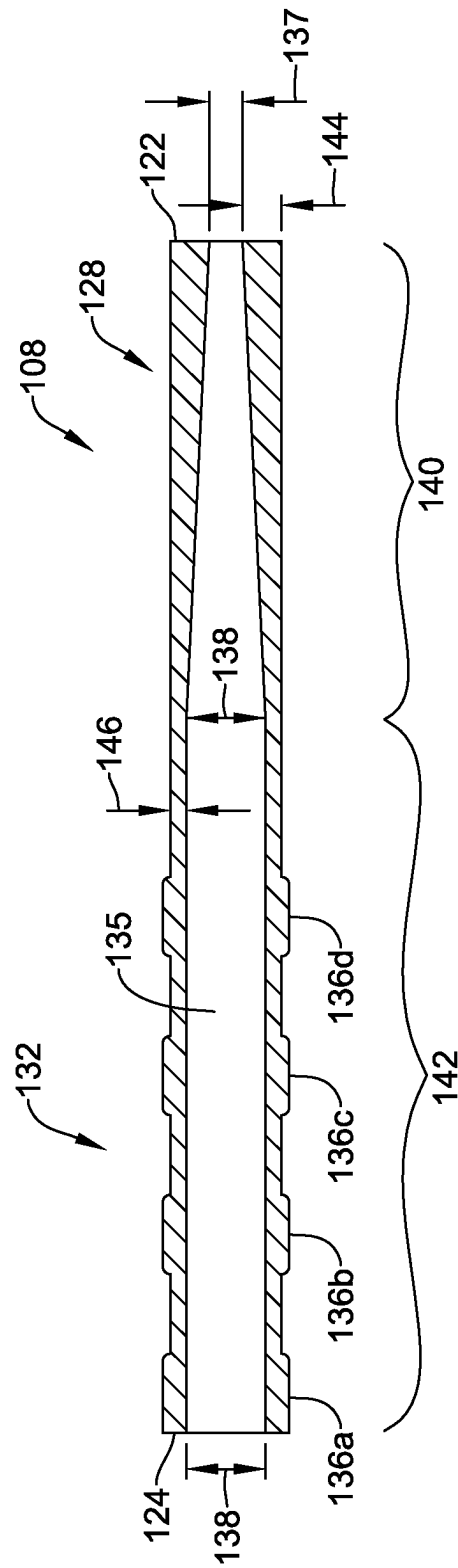


FIG. 3

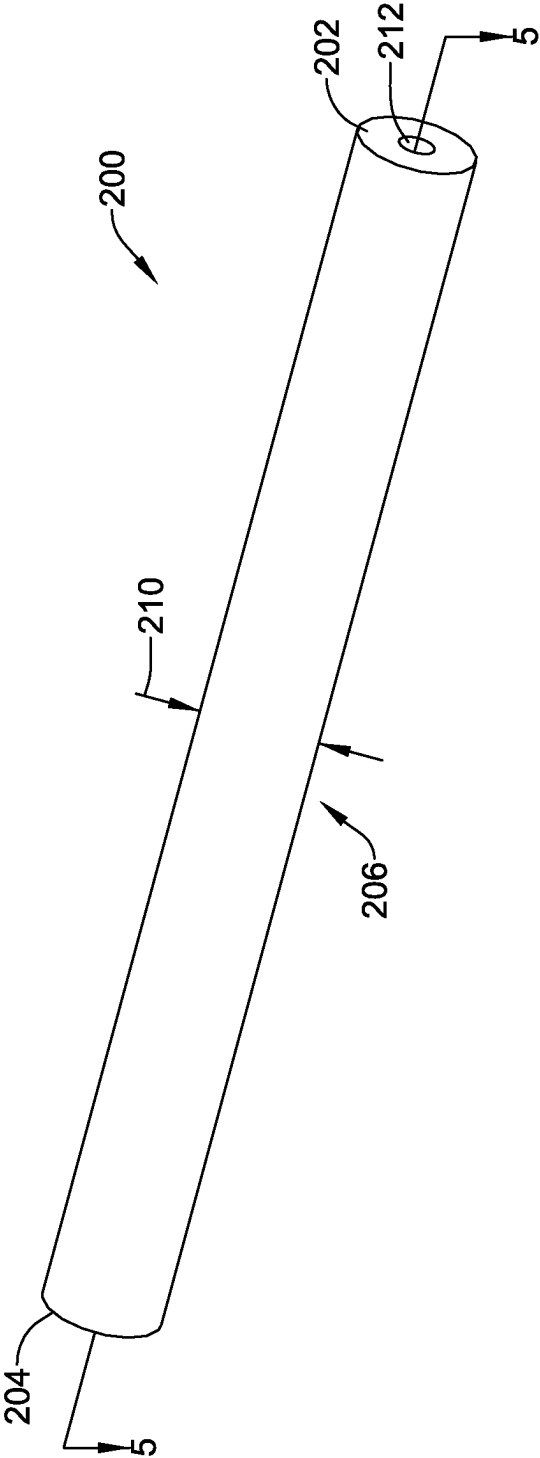


FIG. 4

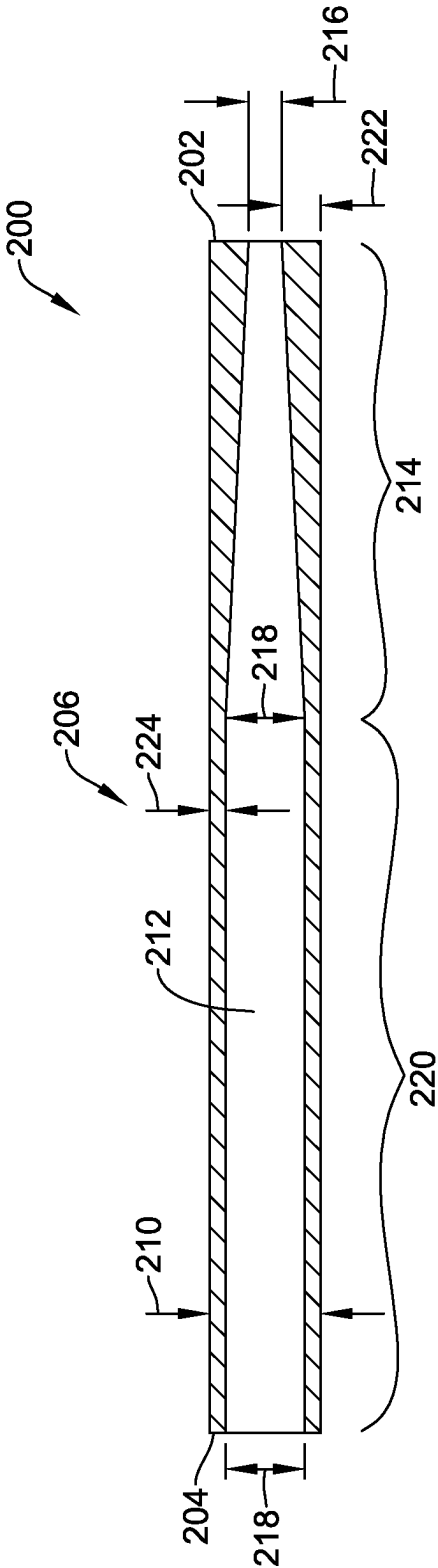
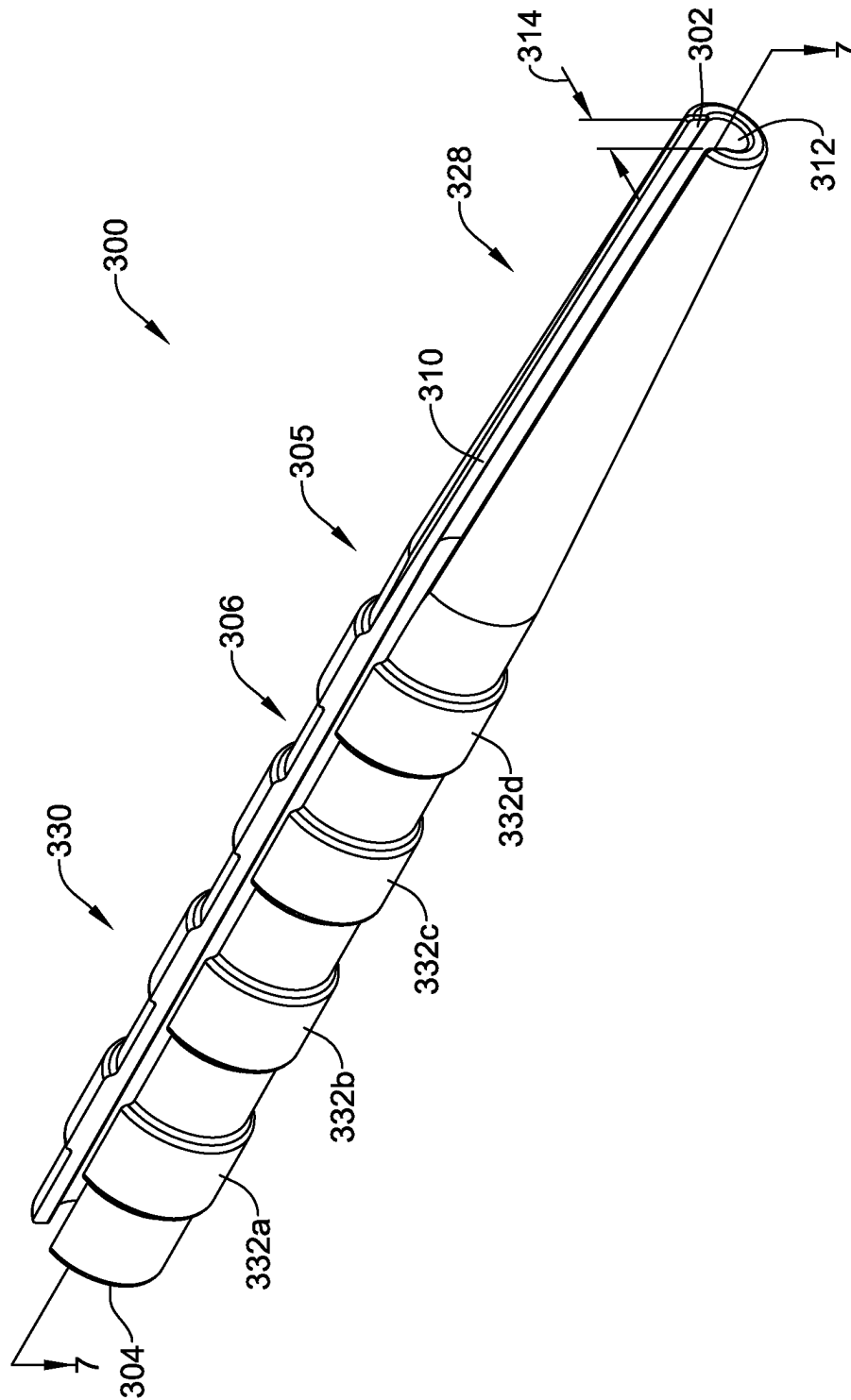


FIG. 5



**FIG. 6**

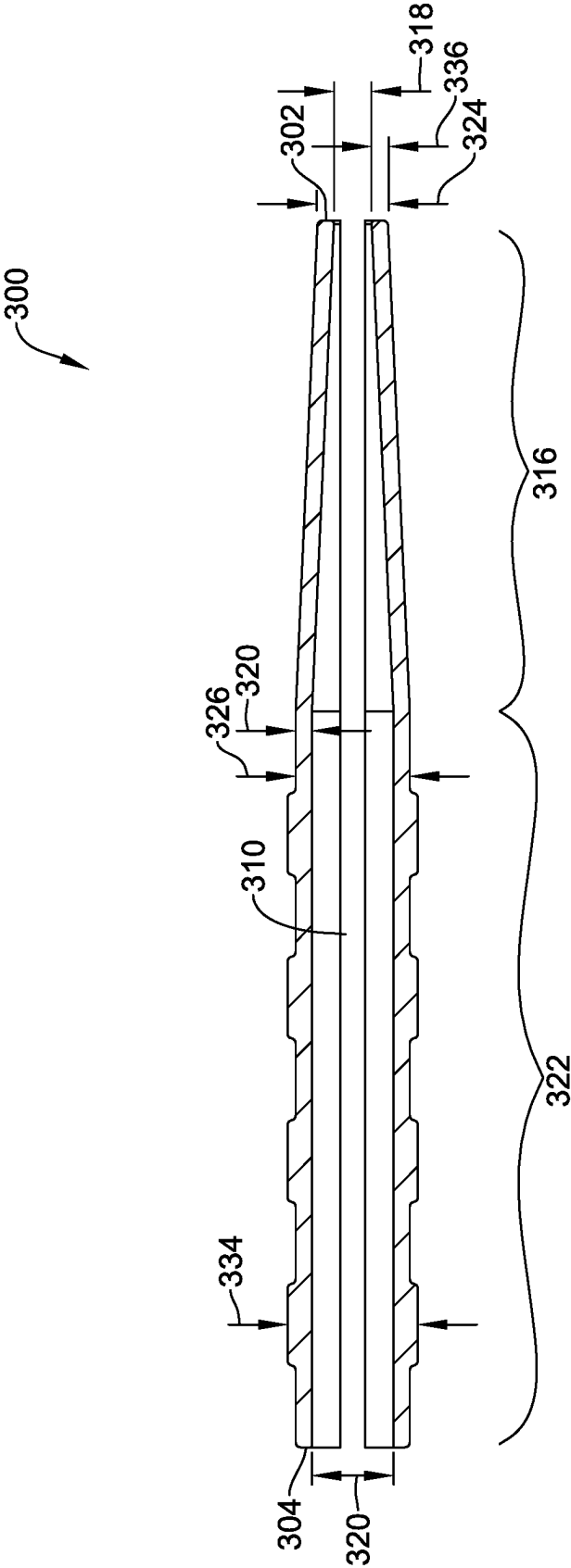


FIG. 7

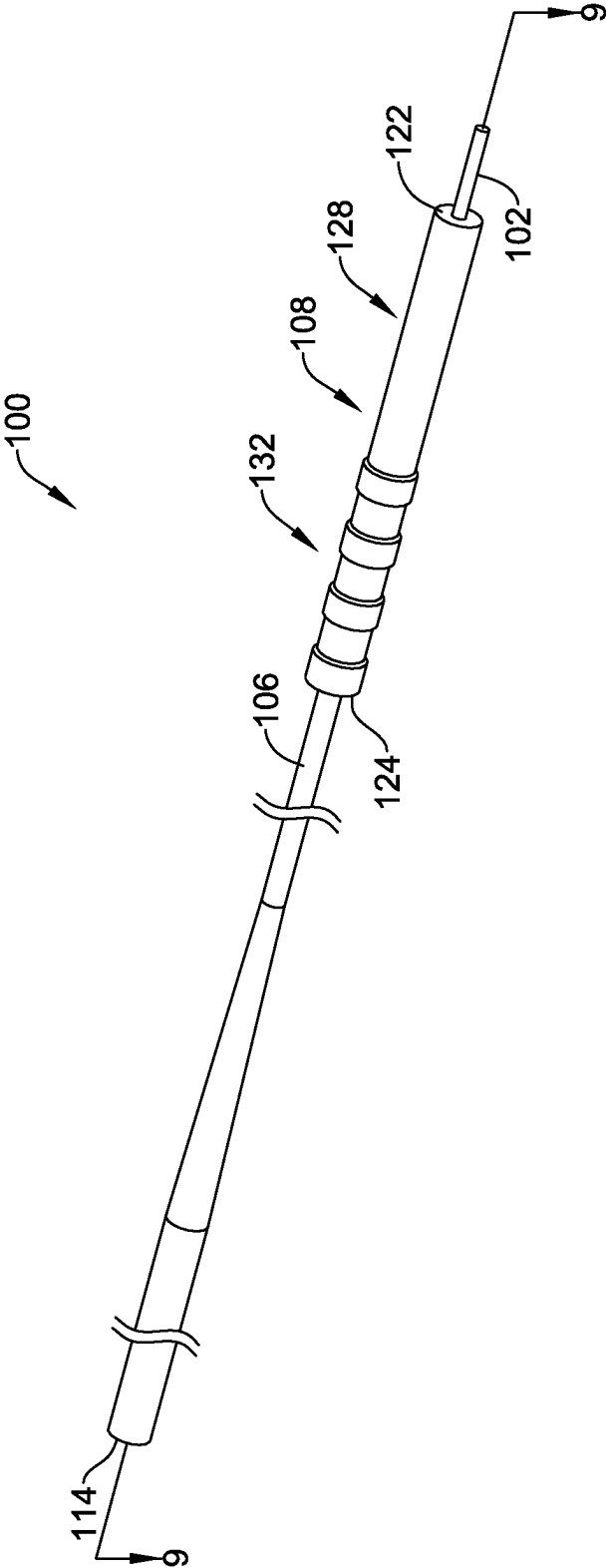


FIG. 8

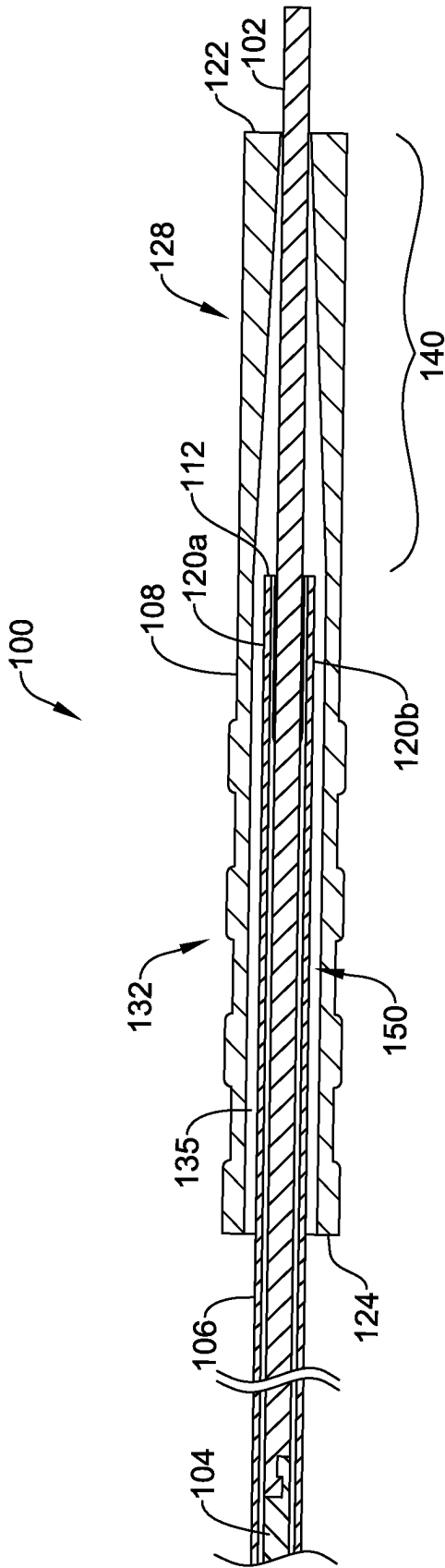


FIG. 9

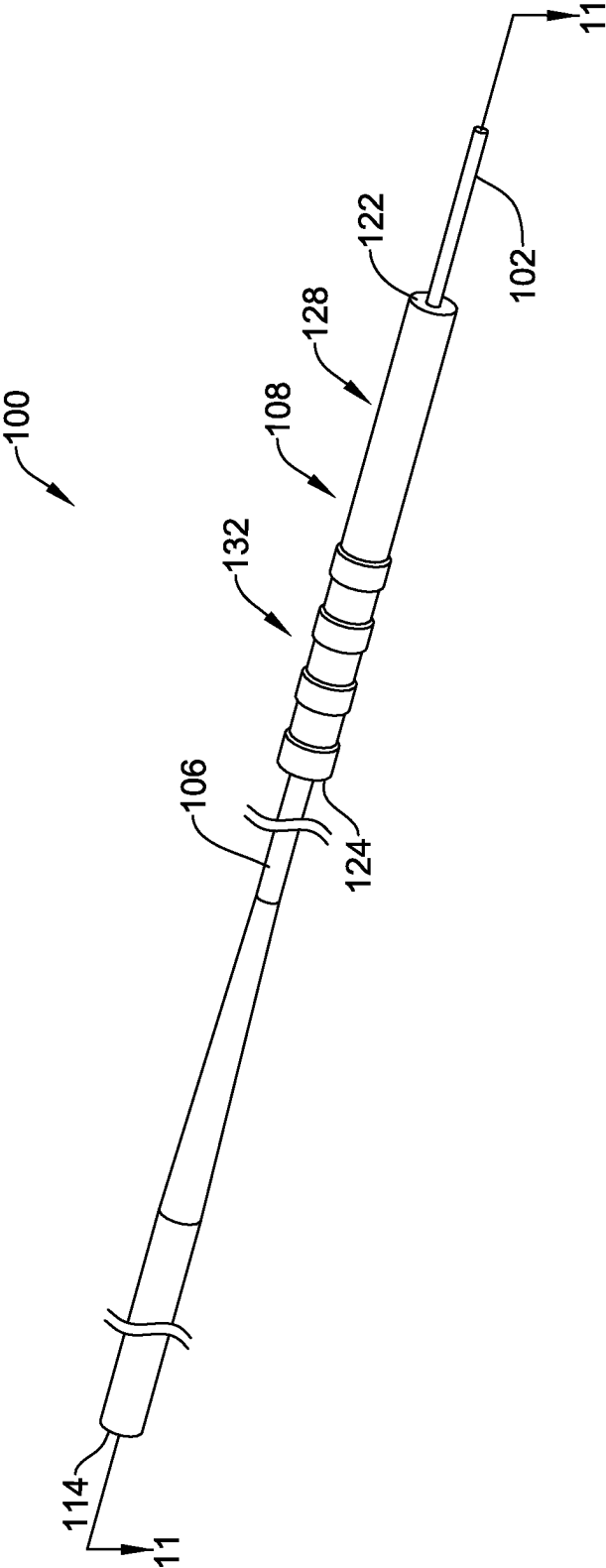


FIG. 10

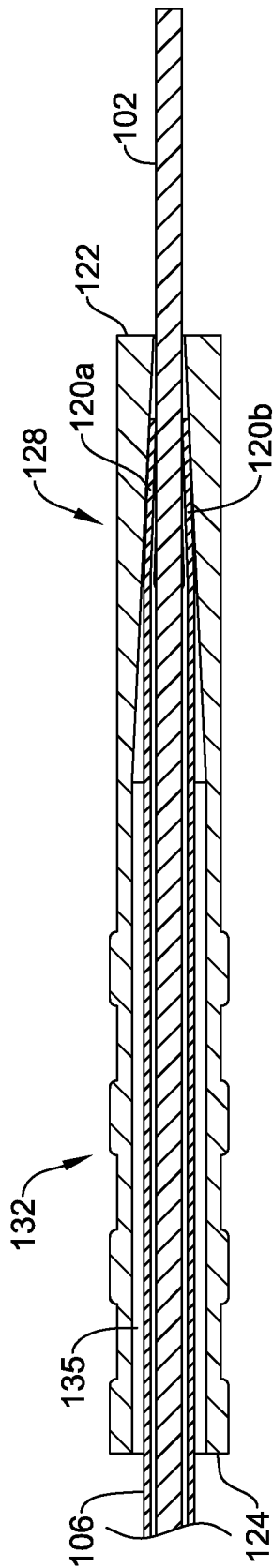


FIG. 11

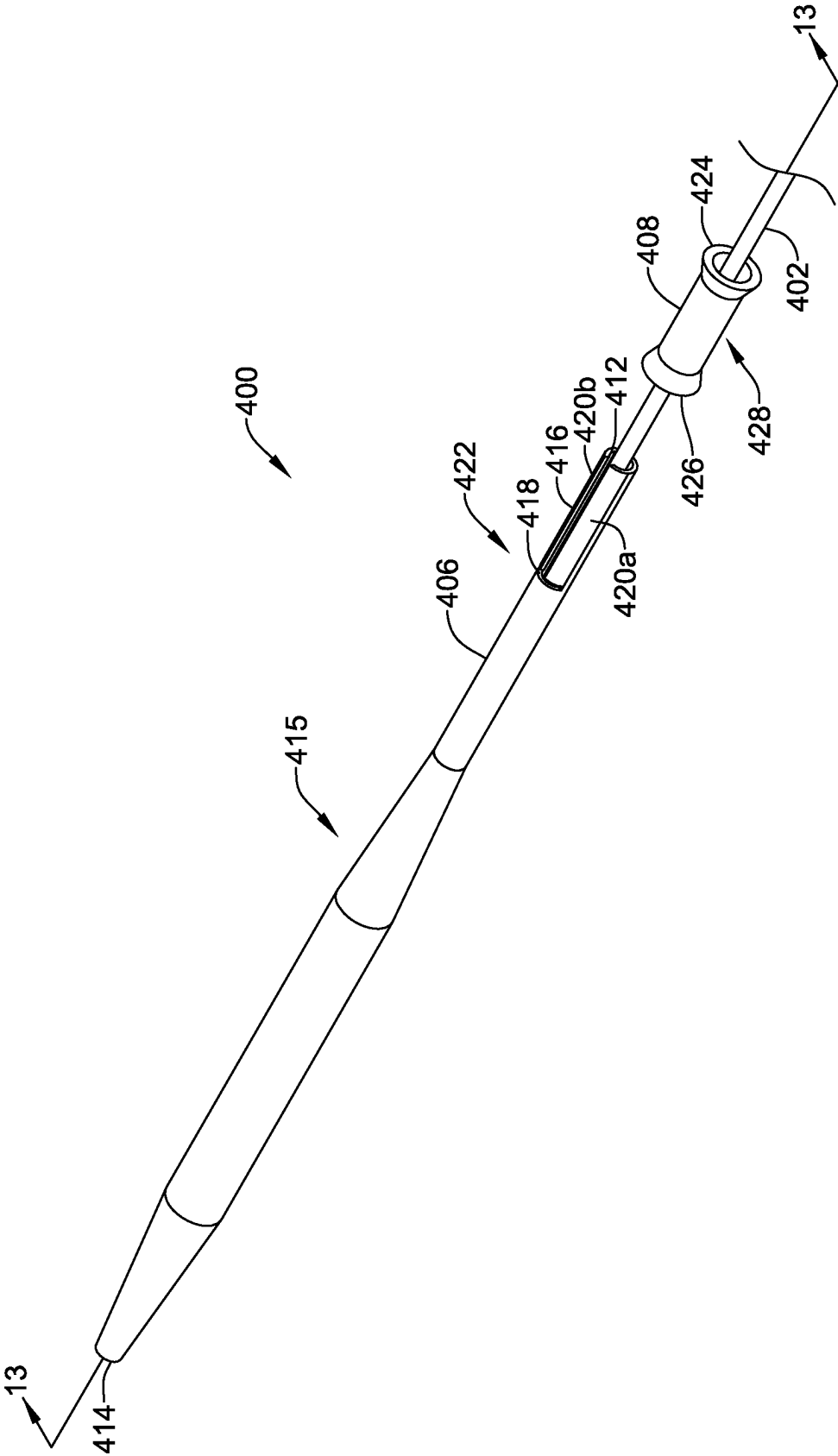


FIG. 12

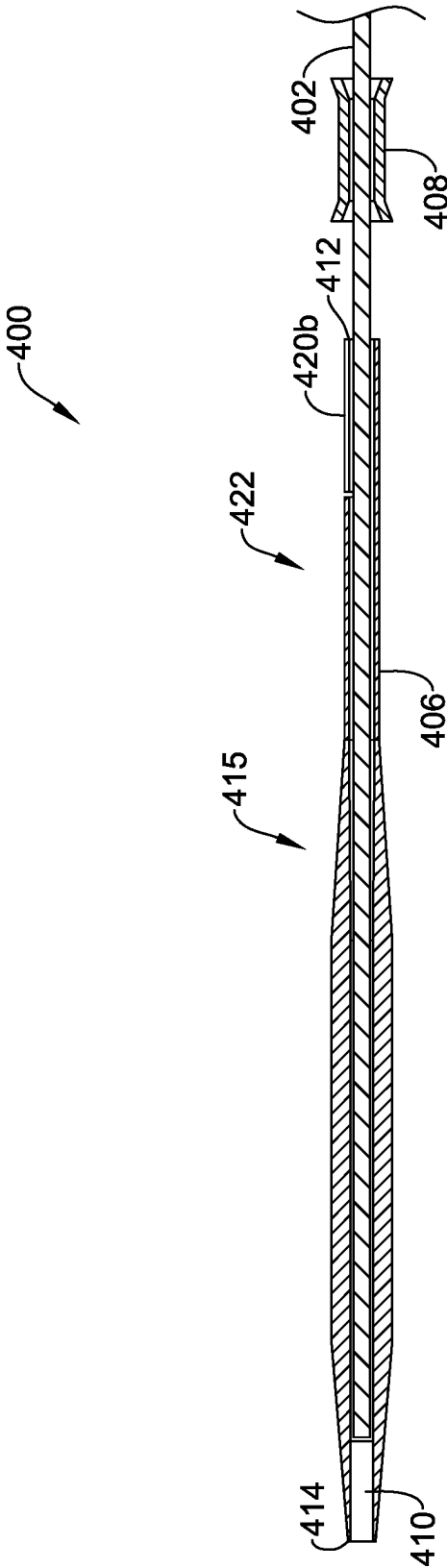


FIG. 13

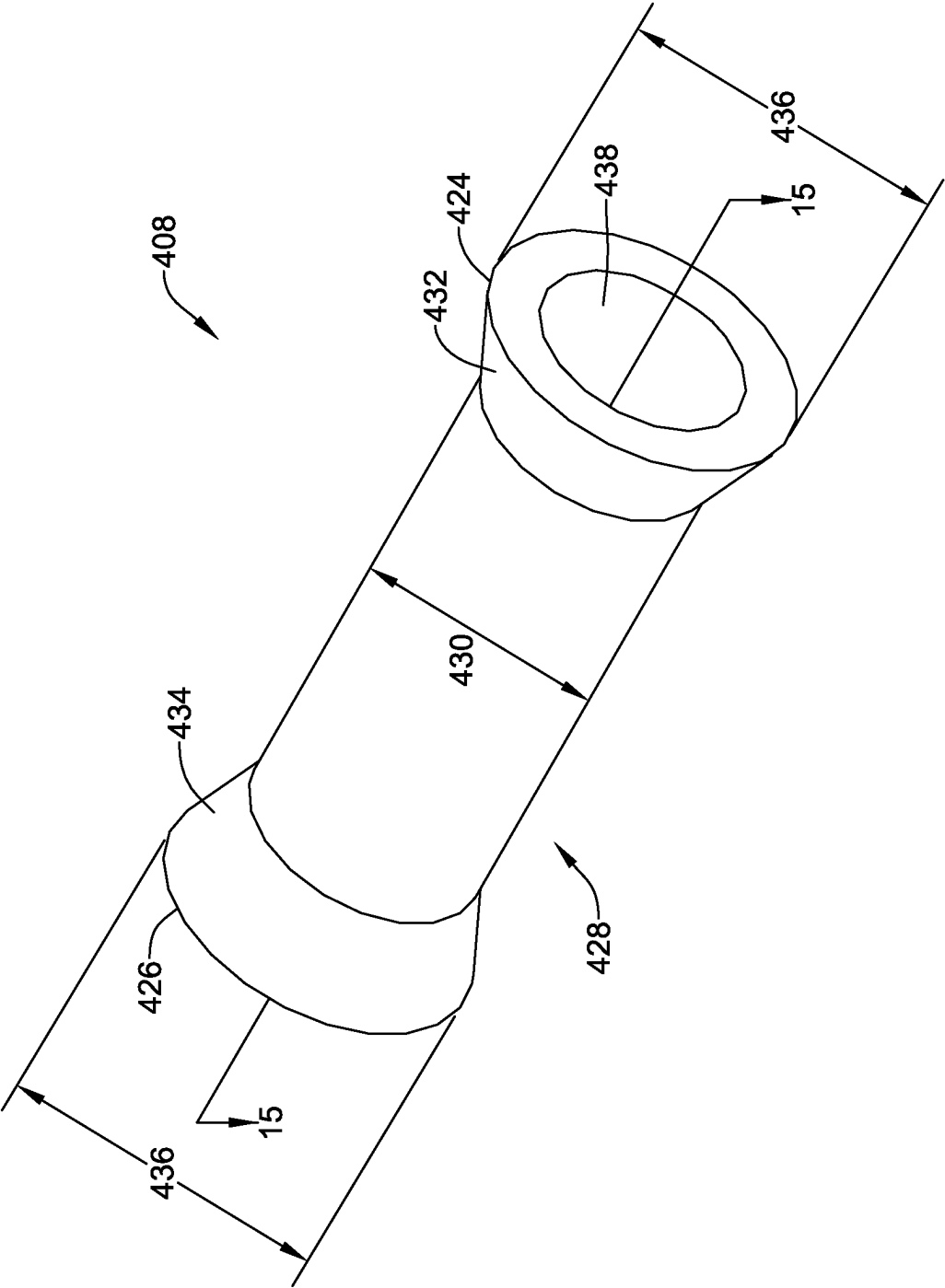


FIG. 14

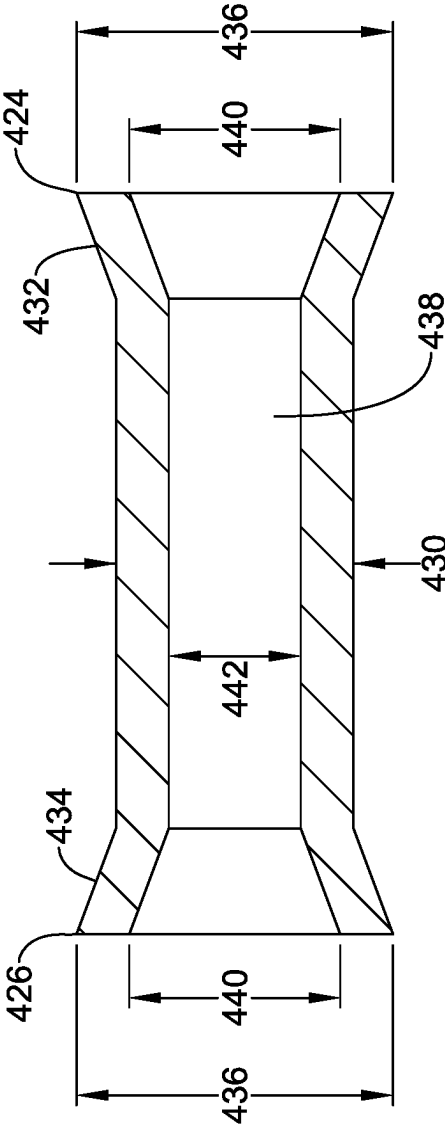


FIG. 15

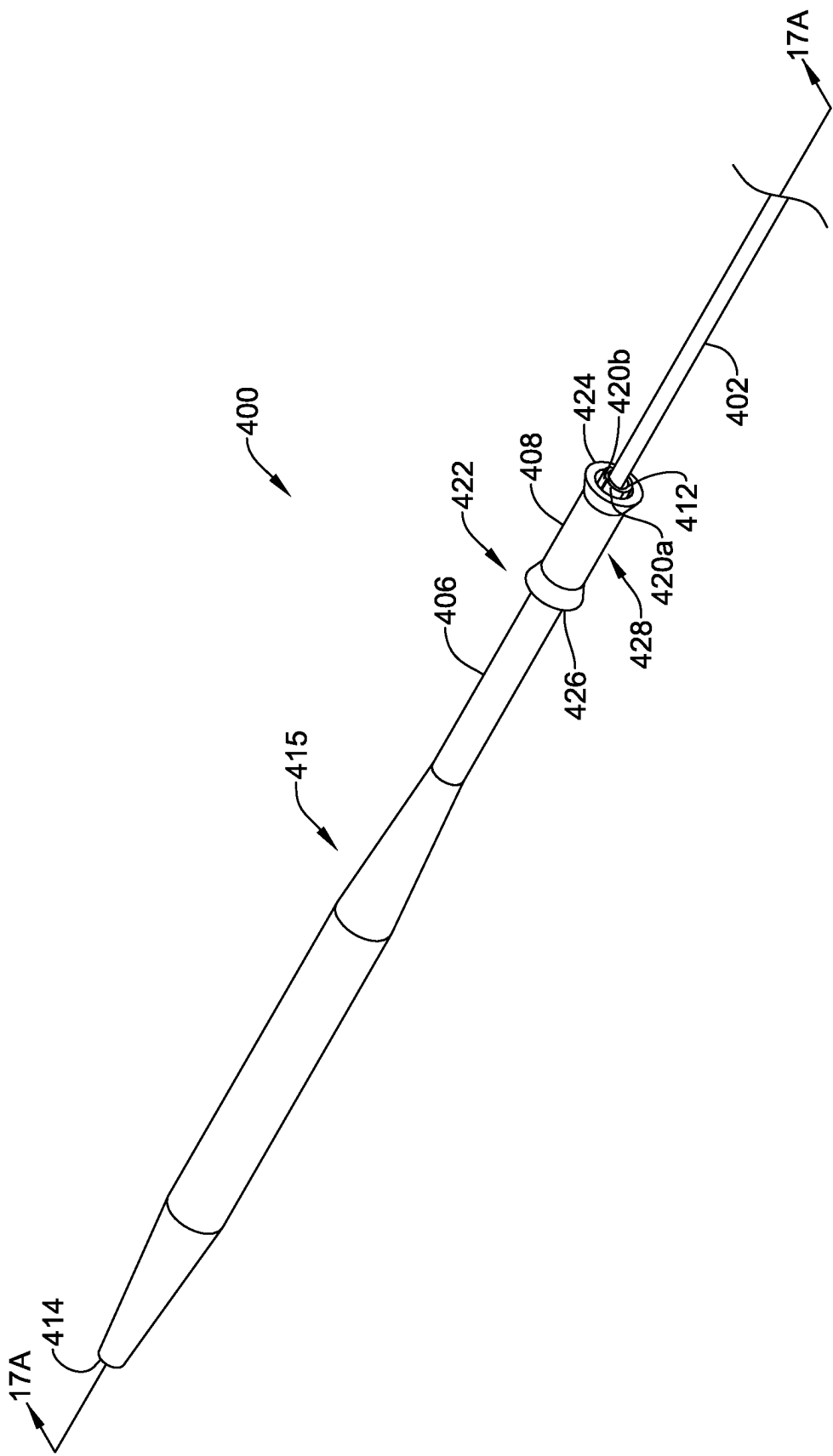


FIG. 16

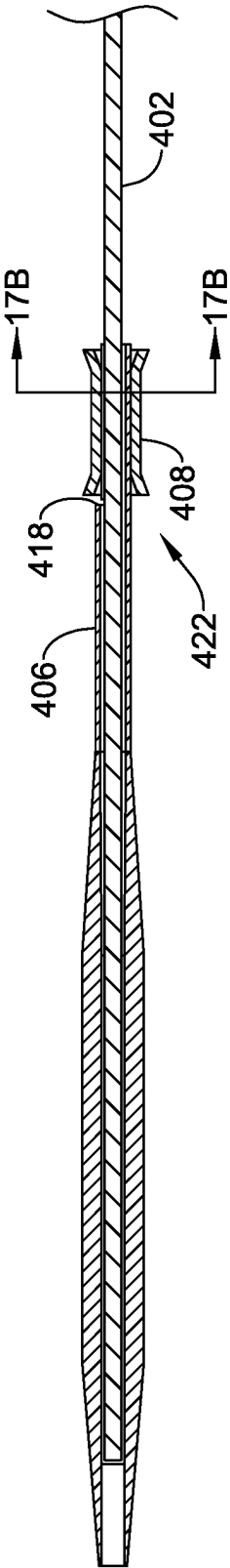


FIG. 17A

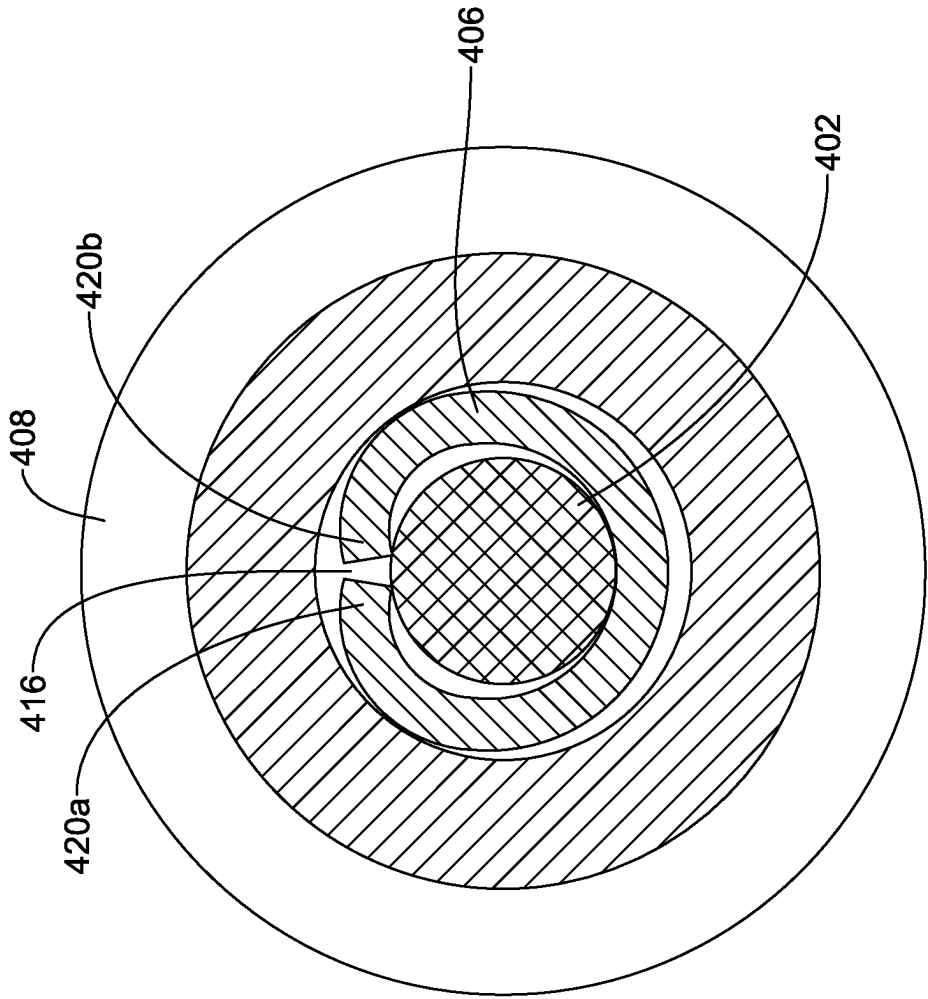


FIG. 17B

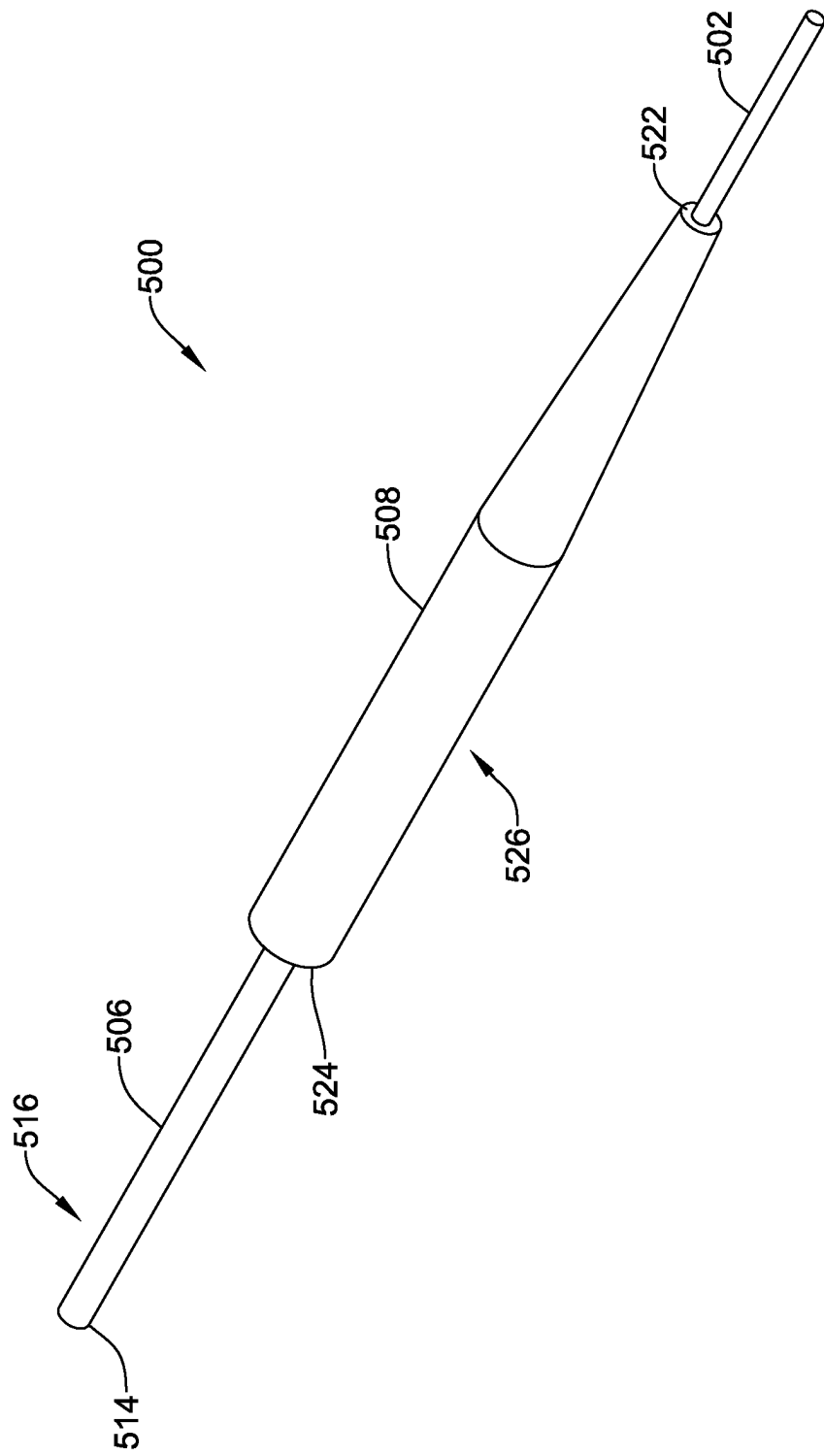
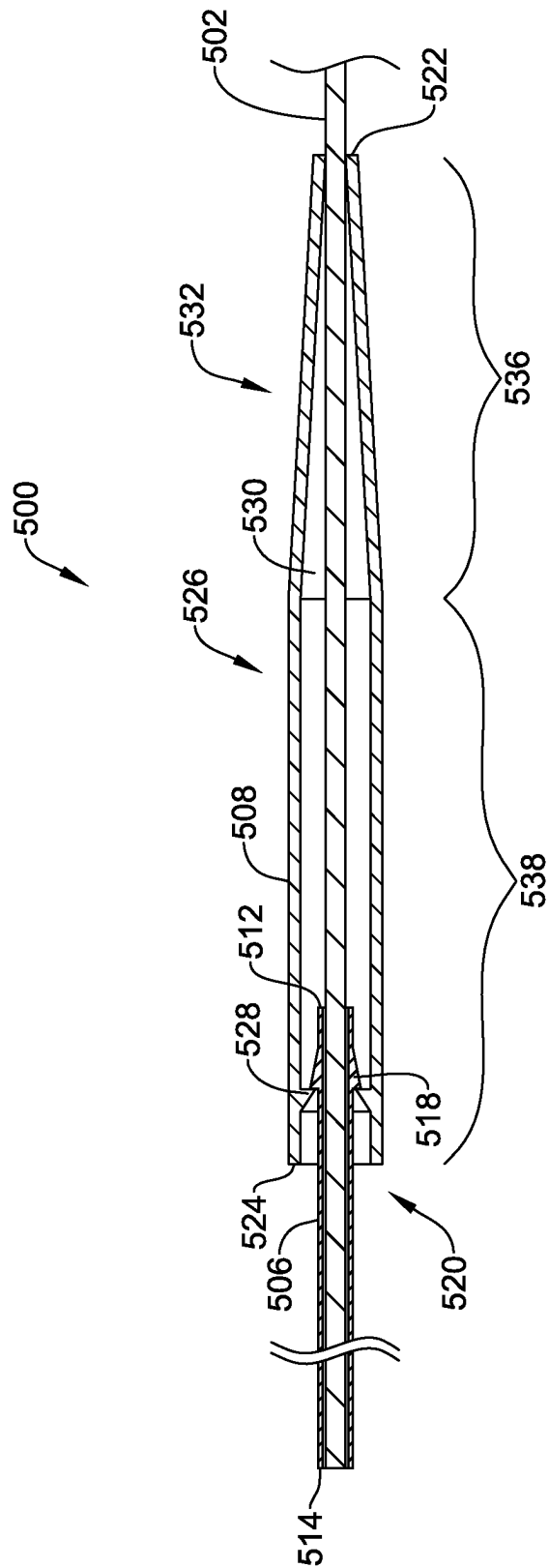


FIG. 18



**FIG. 19**

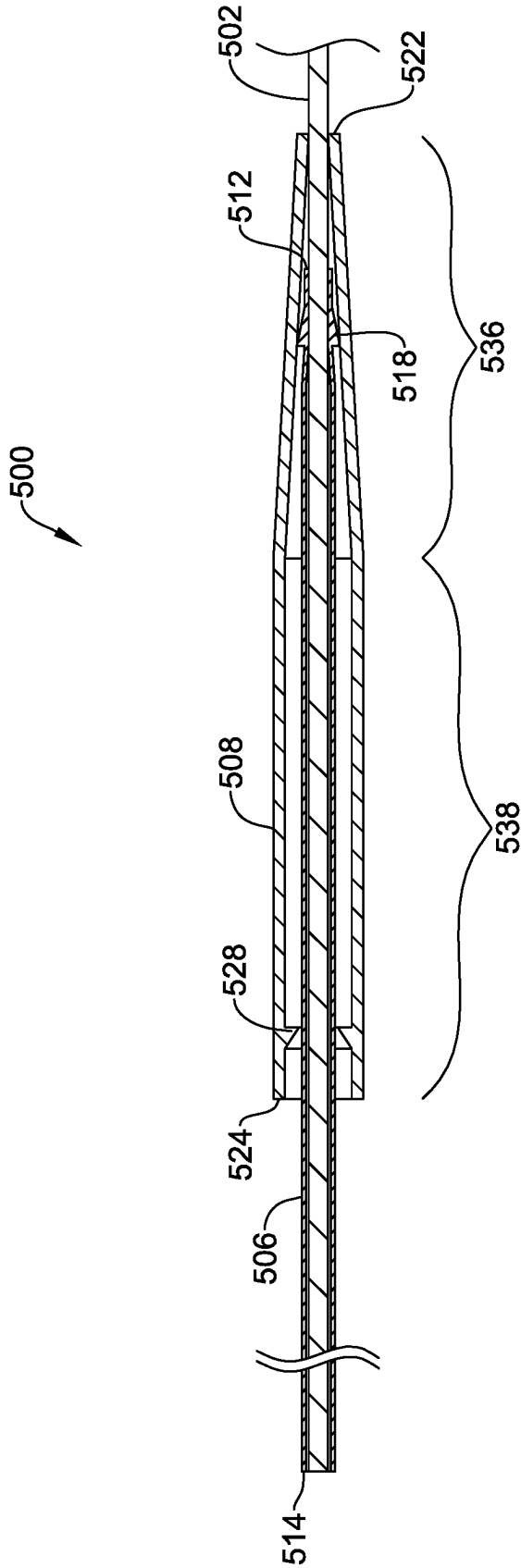


FIG. 20

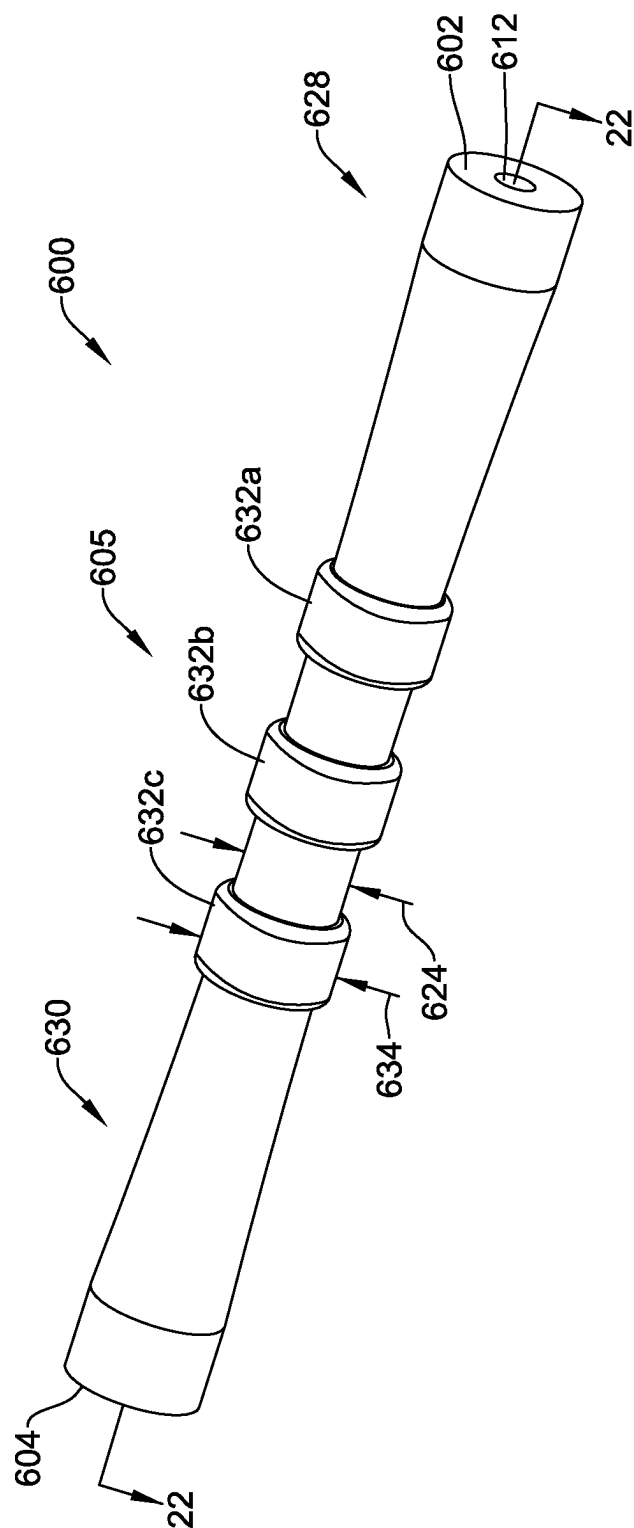


FIG. 21

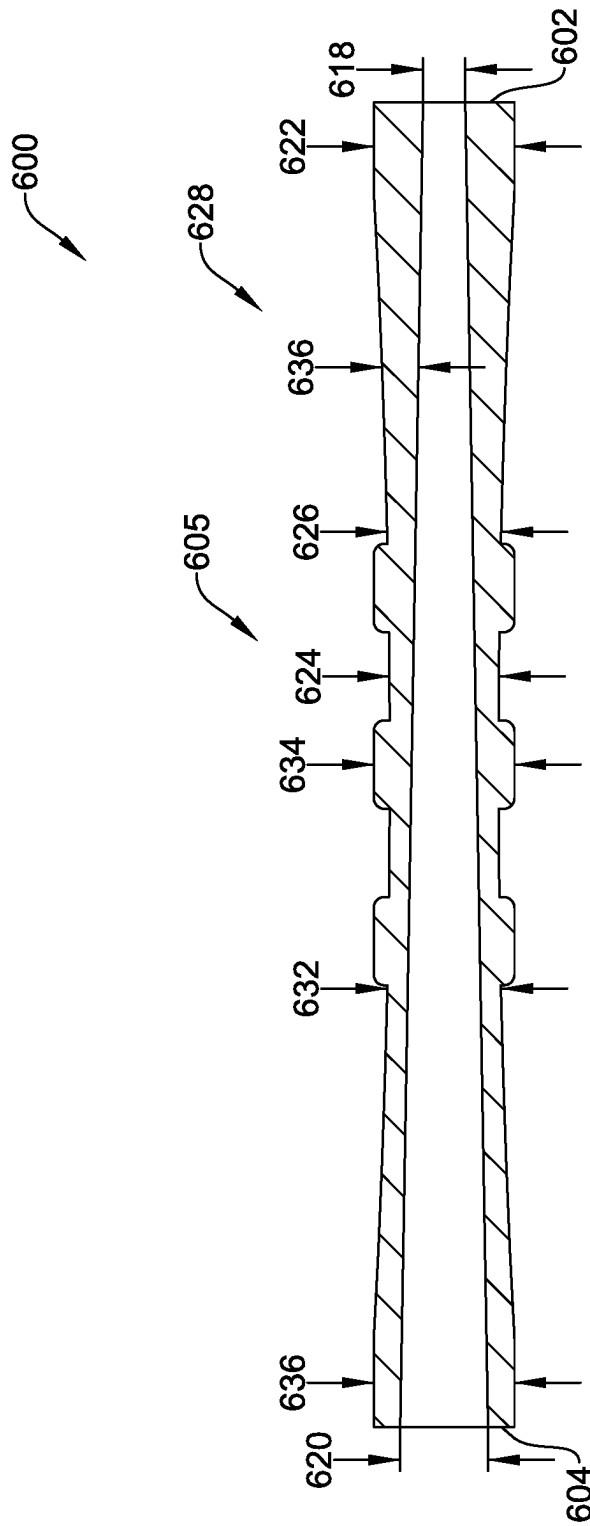


FIG. 22

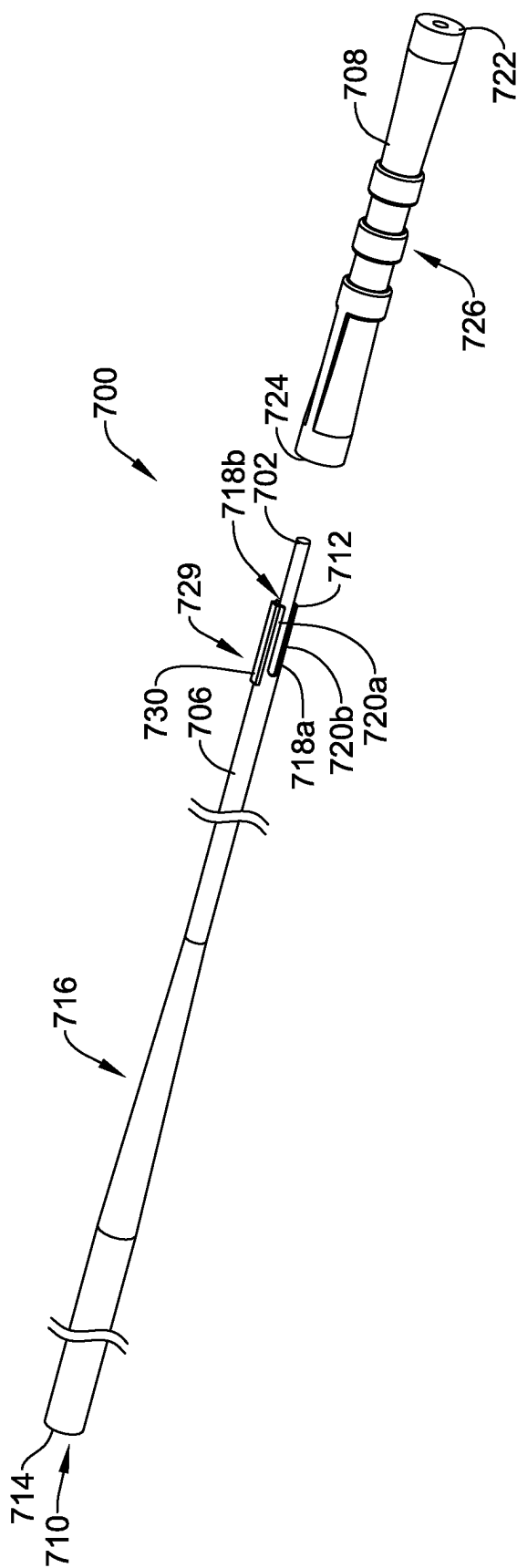


FIG. 23

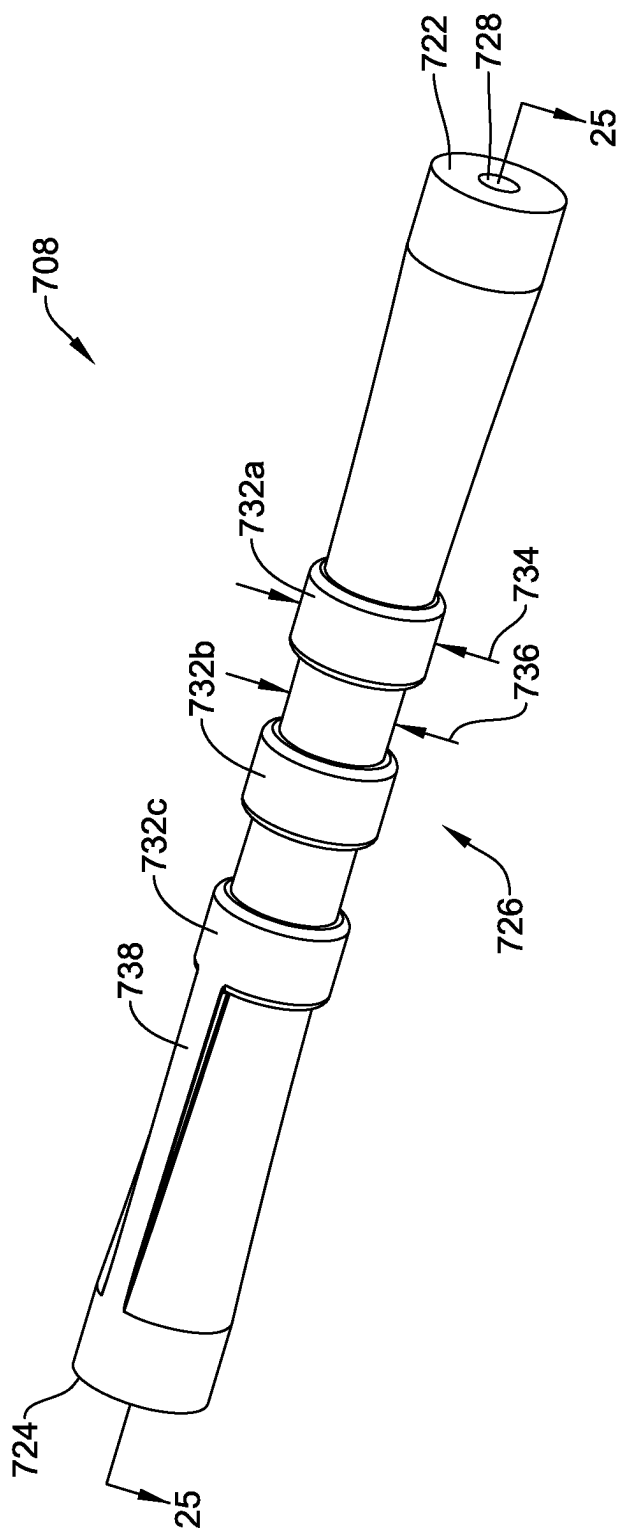


FIG. 24

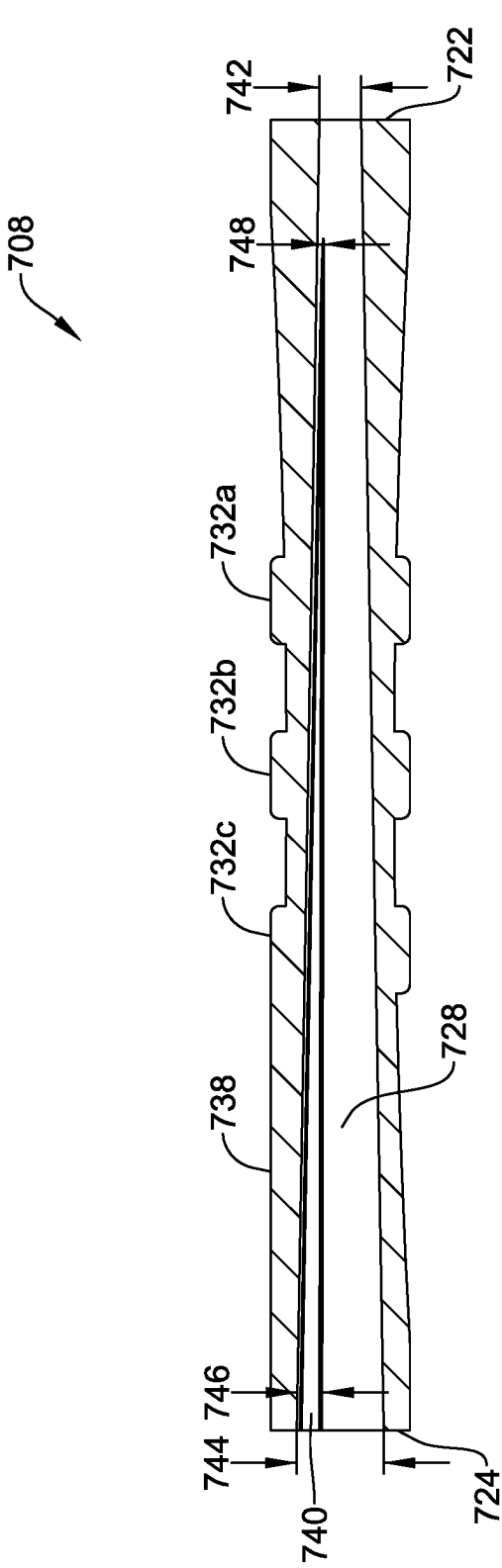


FIG. 25

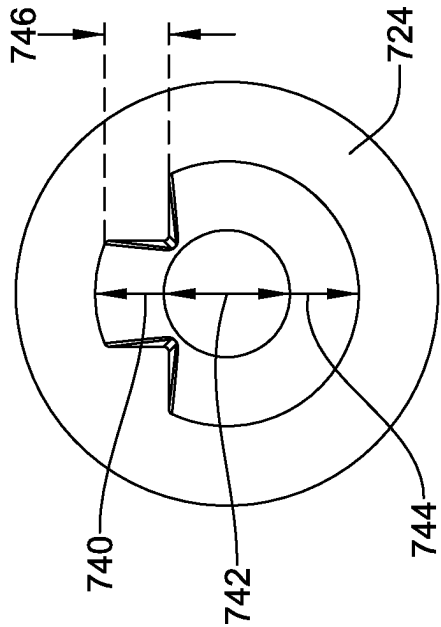


FIG. 26

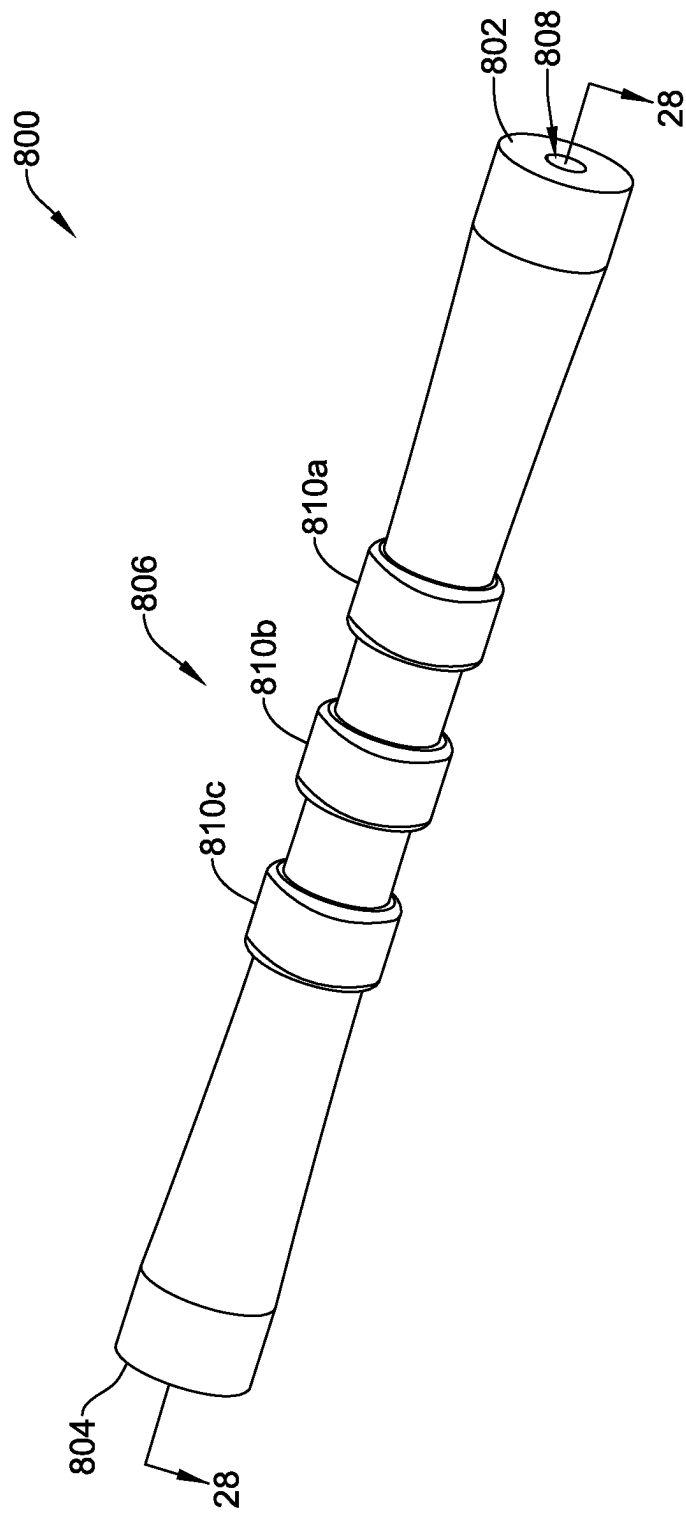


FIG. 27

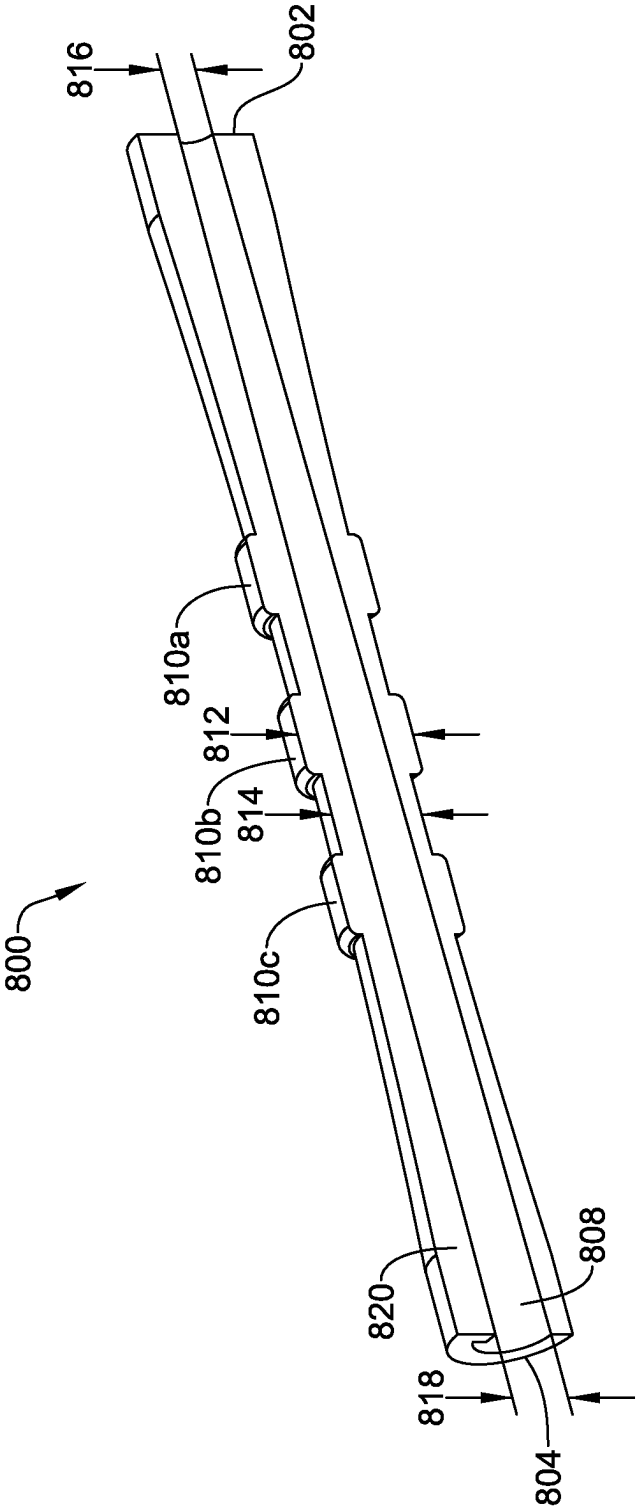


FIG. 28

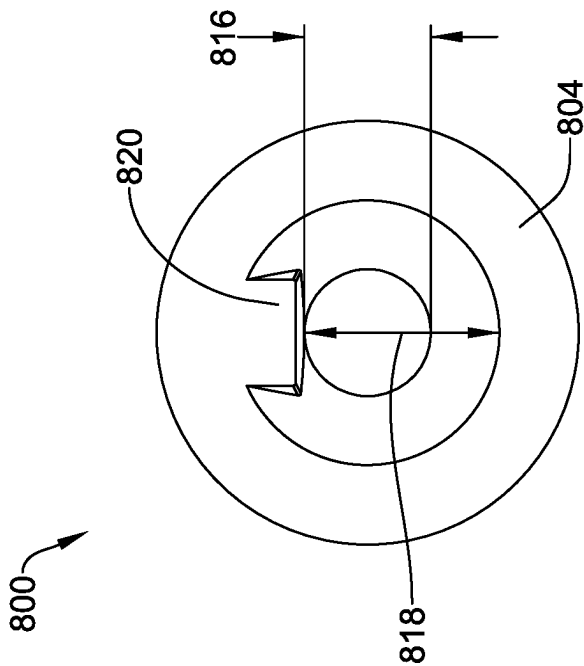


FIG. 29

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## WEDGE-LOCK SHEATH RETENTION MECHANISM

### CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of priority of U.S. Provisional Application No. 62/904,864 filed Sep. 24, 2019, the entire disclosure of which is hereby incorporated by reference.

### TECHNICAL FIELD

The present disclosure pertains to medical devices and methods for manufacturing and/or using medical devices. More particularly, the present disclosure pertains to configurations of a system for securing a medical device within a sheath.

### BACKGROUND

A wide variety of intracorporeal medical devices have been developed for medical use, for example, surgical and/or intravascular use. Some of these devices include guidewires, catheters, medical device delivery systems (e.g., for stents, grafts, replacement valves, etc.), and the like. These devices are manufactured by any one of a variety of different manufacturing methods and may be used according to any one of a variety of methods. There is an ongoing need to provide alternative medical devices as well as alternative methods for manufacturing and/or using medical devices.

### SUMMARY

In a first example, a medical device system may comprise a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, a pusher wire slidably disposed within the lumen of the sheath, and a locking element having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end. The locking element may have a first inner diameter adjacent to the distal end and a second inner diameter adjacent to the proximal end, the second inner diameter smaller than the first inner diameter. A distal end region of the locking element may be configured to freely slide over the sheath. When a proximal end region of the locking element is disposed over the sheath, the locking element may be configured to depress the sheath radially inwards.

Alternatively or additionally to any of the examples above, in another example, the lumen of the locking element within the proximal end region may reduce in diameter in a sloped manner towards the proximal end of the locking element.

Alternatively or additionally to any of the examples above, in another example, the lumen of the locking element within the distal end region may have a generally uniform inner diameter.

Alternatively or additionally to any of the examples above, in another example, the sheath may include a first slot and a second slot, the first and second slots extending distally from the proximal end of the sheath and extending less than an entire length of the sheath.

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Alternatively or additionally to any of the examples above, in another example, the sheath may include a longitudinally extending slot extending distally from the proximal end of the sheath.

Alternatively or additionally to any of the examples above, in another example, the longitudinally extending slot may terminate at a circumferentially extending slit.

Alternatively or additionally to any of the examples above, in another example, the sheath may include an angled protrusion extending radially outward from an outer surface thereof.

Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the sheath may reduce in diameter in a proximal direction.

Alternatively or additionally to any of the examples above, in another example, the locking element may include an angled protrusion extending radially inward from an inner surface of the lumen of the locking element.

Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the locking element may reduce in diameter in a distal direction.

Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the sheath and the angled protrusion of the locking element may be configured to cooperate to limit axial movement of the locking element.

Alternatively or additionally to any of the examples above, in another example, the locking element may further comprise a slot extending from a proximal end of the locking element to a distal end of the locking element, the slot extending through a thickness of a wall of the locking element.

Alternatively or additionally to any of the examples above, in another example, the slot may have a width that is greater than a width of the pusher wire and less than a width of the sheath.

Alternatively or additionally to any of the examples above, in another example, the locking element may comprise a plurality of circumferentially extending ribs extending from an outer surface thereof.

Alternatively or additionally to any of the examples above, in another example, the locking element may be formed from a more rigid material than the sheath.

In another example, a medical device system may comprise a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the sheath having a first slot and a second slot, the first and second slots extending distally from the proximal end of the sheath, extending less than an entire length of the sheath, and defining a first flexible arm and a second flexible arm, a pusher wire slidably disposed within the lumen of the sheath, and a locking element having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the locking element having a first inner diameter adjacent to the distal end and a second inner diameter adjacent to the proximal end, the second inner diameter smaller than the first inner diameter. A distal end region of the locking element may be configured to freely slide over the sheath. When a proximal end region of the locking element is disposed over a proximal end region of the sheath, the locking element may be configured to depress the first and second flexible arms radially inwards.

Alternatively or additionally to any of the examples above, in another example, the lumen of the locking element

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within proximal end region may reduce in diameter in a sloped manner towards the proximal end of the locking element.

Alternatively or additionally to any of the examples above, in another example, the lumen of the locking element within the distal end region may have a generally uniform inner diameter.

Alternatively or additionally to any of the examples above, in another example, the locking element may further comprise a slot extending from a proximal end of the locking element to a distal end of the locking element, the slot extending through a thickness of a wall of the locking element.

Alternatively or additionally to any of the examples above, in another example, the slot may have a width that is greater than a width of the pusher wire and less than a width of the sheath.

Alternatively or additionally to any of the examples above, in another example, the locking element may comprise a plurality of circumferentially extending ribs extending from an outer surface thereof.

Alternatively or additionally to any of the examples above, in another example, the locking element may have a generally uniform wall thickness from the proximal end to the distal end of the locking element.

Alternatively or additionally to any of the examples above, in another example, the locking element may have a first wall thickness adjacent the proximal end and a second wall thickness adjacent to the distal end, the first wall thickness greater than the second wall thickness.

In another example, a medical device system may comprise a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the sheath further comprising a longitudinally extending slot extending distally from the proximal end of the sheath, a pusher wire slidably disposed within the lumen of the sheath, and a locking element having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the lumen of the locking element having a generally hourglass shape. A distal end region of the locking element may be configured to freely slide over the sheath. When an intermediate region of the locking element is disposed over the sheath, the locking element may be configured to depress the sheath radially inwards.

Alternatively or additionally to any of the examples above, in another example, the longitudinally extending slot of the sheath may terminate at a circumferentially extending slit.

Alternatively or additionally to any of the examples above, in another example, the sheath may further comprise a first flap and a second flap adjacent the longitudinally extending slot.

Alternatively or additionally to any of the examples above, in another example, the locking element may have a first inner diameter adjacent the distal end and a second inner diameter adjacent to the intermediate region, the second inner diameter smaller than the first inner diameter.

Alternatively or additionally to any of the examples above, in another example, the second inner diameter may be generally constant and extends between a flared distal end region and a flared proximal end region.

In another example, a medical device system may comprise a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and

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the distal end, and a lumen extending from the proximal end to the distal end, the sheath further comprising an angled protrusion extending radially outward from an outer surface thereof, a pusher wire slidably disposed within the lumen of the sheath, and a locking element. The locking element may comprise a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, a first inner diameter adjacent to the distal end, a second inner diameter adjacent to the proximal end, the second inner diameter smaller than the first outer diameter, and an angled protrusion extending radially inward from an inner surface of the lumen of the locking element. A distal end region of the locking element may be configured to freely slide over the sheath. When a proximal end region of the locking element is disposed over the sheath, the locking element may be configured to depress the sheath radially inwards.

Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the sheath may reduce in diameter in a proximal direction.

Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the sheath may be positioned adjacent to the proximal end of the sheath.

Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the locking element may reduce in diameter in a distal direction.

Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the locking element may be positioned adjacent to the distal end of the locking element.

Alternatively or additionally to any of the examples above, in another example, the angled protrusion of the sheath and the angled protrusion of the locking element may be configured to cooperate to limit axial movement of the locking element.

Alternatively or additionally to any of the examples above, in another example, the lumen of the locking element within the proximal end region may reduce in diameter in a sloped manner towards the proximal end of the locking element.

The above summary of some embodiments, aspects, and/or examples is not intended to describe each embodiment or every implementation of the present disclosure. The figures and the detailed description which follows more particularly exemplify these embodiments.

#### BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure may be more completely understood in consideration of the following detailed description of various embodiments in connection with the accompanying drawings, in which:

FIG. 1 is a perspective view of an example medical device system in a first configuration;

FIG. 2 is a perspective view of an illustrative locking mechanism;

FIG. 3 is a cross-sectional view of the illustrative locking mechanism of FIG. 2;

FIG. 4 is a perspective view of another illustrative locking mechanism;

FIG. 5 is a cross-sectional view of the illustrative locking mechanism of FIG. 4;

FIG. 6 is a perspective view of another illustrative locking mechanism;

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FIG. 7 is a cross-sectional view of the illustrative locking mechanism of FIG. 6;

FIG. 8 is a perspective view of the illustrative medical device system of FIG. 1 in a second configuration;

FIG. 9 is a partial cross-sectional view of the illustrative medical device system of FIG. 8;

FIG. 10 is a perspective view of the illustrative medical device system of FIGS. 1 and 8 in a third configuration;

FIG. 11 is a partial cross-sectional view of the illustrative medical device system of FIG. 10;

FIG. 12 is a perspective view of another example medical device system in a first configuration;

FIG. 13 is a partial cross-sectional view of the illustrative medical device system of FIG. 12;

FIG. 14 is a perspective view of another illustrative locking mechanism;

FIG. 15 is a cross-sectional view of the illustrative locking mechanism of FIG. 14;

FIG. 16 is a perspective view of the illustrative medical device system of FIG. 12 in a second configuration;

FIG. 17A is a partial cross-sectional view of the illustrative medical device system of FIG. 16;

FIG. 17B is a partial cross-sectional view of the illustrative medical device system of FIG. 17A;

FIG. 18 is a perspective view of another example medical device system in a first configuration;

FIG. 19 is a partial cross-sectional view of the illustrative medical device system of FIG. 18;

FIG. 20 is a partial cross-sectional view of the illustrative medical device system of FIG. 18 in a second configuration;

FIG. 21 is a perspective view of another illustrative locking mechanism;

FIG. 22 is a cross-sectional view of the illustrative locking mechanism of FIG. 21;

FIG. 23 is a perspective view of another example medical device system in a first configuration;

FIG. 24 is a perspective view of an illustrative locking mechanism;

FIG. 25 is a cross-sectional view of the illustrative locking mechanism of FIG. 24;

FIG. 26 is a distal end view of the illustrative locking mechanism of FIG. 24;

FIG. 27 is a perspective view of an illustrative locking mechanism;

FIG. 28 is a perspective cross-sectional view of the illustrative locking mechanism of FIG. 27; and

FIG. 29 is a distal end view of the illustrative locking mechanism of FIG. 27.

While aspects of the disclosure are amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit aspects of the disclosure to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.

#### DETAILED DESCRIPTION

The following description should be read with reference to the drawings, which are not necessarily to scale, wherein like reference numerals indicate like elements throughout the several views. The detailed description and drawings are intended to illustrate but not limit the claimed invention. Those skilled in the art will recognize that the various elements described and/or shown may be arranged in vari-

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ous combinations and configurations without departing from the scope of the disclosure. The detailed description and drawings illustrate example embodiments of the claimed invention. However, in the interest of clarity and ease of understanding, while every feature and/or element may not be shown in each drawing, the feature(s) and/or element(s) may be understood to be present regardless, unless otherwise specified.

For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.

All numeric values are herein assumed to be modified by the term “about,” whether or not explicitly indicated. The term “about,” in the context of numeric values, generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (e.g., having the same function or result). In many instances, the term “about” may include numbers that are rounded to the nearest significant figure. Other uses of the term “about” (e.g., in a context other than numeric values) may be assumed to have their ordinary and customary definition(s), as understood from and consistent with the context of the specification, unless otherwise specified.

The recitation of numerical ranges by endpoints includes all numbers within that range, including the endpoints (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).

Although some suitable dimensions, ranges, and/or values pertaining to various components, features and/or specifications are disclosed, one of skill in the art, incited by the present disclosure, would understand desired dimensions, ranges, and/or values may deviate from those expressly disclosed.

As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise. It is to be noted that in order to facilitate understanding, certain features of the disclosure may be described in the singular, even though those features may be plural or recurring within the disclosed embodiment(s). Each instance of the features may include and/or be encompassed by the singular disclosure(s), unless expressly stated to the contrary. For simplicity and clarity purposes, not all elements of the disclosed invention are necessarily shown in each figure or discussed in detail below. However, it will be understood that the following discussion may apply equally to any and/or all of the components for which there are more than one, unless explicitly stated to the contrary. Additionally, not all instances of some elements or features may be shown in each figure for clarity.

Relative terms such as “proximal,” “distal,” “advance,” “retract,” variants thereof, and the like, may be generally considered with respect to the positioning, direction, and/or operation of various elements relative to a user/operator/manipulator of the device, wherein “proximal” and “retract” indicate or refer to closer to or toward the user and “distal” and “advance” indicate or refer to farther from or away from the user. In some instances, the terms “proximal” and “distal” may be arbitrarily assigned in an effort to facilitate understanding of the disclosure, and such instances will be readily apparent to the skilled artisan. Other relative terms, such as “upstream,” “downstream,” “inflow,” and “outflow” refer to a direction of fluid flow within a lumen, such as a body lumen, a blood vessel, or within a device. Still other relative terms, such as “axial,” “circumferential,” “longitu-

dinal”, “lateral”, “radial”, etc. and/or variants thereof generally refer to direction and/or orientation relative to a central longitudinal axis of the disclosed structure or device.

It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment(s) described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect the particular feature, structure, or characteristic in connection with other embodiments, whether or not explicitly described, unless clearly stated to the contrary. That is, the various individual elements described below, even if not explicitly shown in a particular combination, are nevertheless contemplated as being combinable or arrangeable with each other to form other additional embodiments or to complement and/or enrich the described embodiment(s), as would be understood by one of ordinary skill in the art.

For the purpose of clarity, certain identifying numerical nomenclature (e.g., first, second, third, fourth, etc.) may be used throughout the description and/or claims to name and/or differentiate between various described and/or claimed features. It is to be understood that the numerical nomenclature is not intended to be limiting and is exemplary only. In some embodiments, alterations of and deviations from previously-used numerical nomenclature may be made in the interest of brevity and clarity. That is, a feature identified as a “first” element may later be referred to as a “second” element, a “third” element, etc. or may be omitted entirely, and/or a different feature may be referred to as the “first” element. The meaning and/or designation in each instance will be apparent to the skilled practitioner.

Diseases and/or medical conditions that impact and/or are affected by the cardiovascular system are prevalent throughout the world. For example, some forms of arterial venous malformations (AVMs) may “feed” off of normal blood flow through the vascular system. Without being bound by theory, it is believed that it may be possible to treat, at least partially, arterial venous malformations and/or other diseases or conditions by starving them of normal, oxygen and/or nutrient-rich blood flow, thereby limiting their ability to grow and/or spread. Other examples of diseases or conditions that may benefit from vascular occlusion include, but are not limited to, bleeds, aneurysms, venous insufficiency, shutting off blood flow prior to organ resection, or preventing embolic bead reflux into branch vessels in the liver. Disclosed herein are medical devices that may be used within a portion of the cardiovascular system in order to treat and/or repair some arterial venous malformations and/or other diseases or conditions. The devices disclosed herein may also provide a number of additional desirable features and benefits as described in more detail below.

FIG. 1 is a perspective view of an example medical device system **100** in a partially unassembled configuration. The medical device system **100** may include a pusher wire **102**, an implant **104** (see, for example, FIG. 9), such as, but not limited to, an embolic coil, an introducer sheath **106**, and a locking element **108**. For simplicity, the implant **104** is described as an embolic coil, but other suitable medical devices transported, delivered, used, released, etc. in a similar manner are also contemplated, including but not limited to, vascular occlusion devices coils, stents, embolic

filters, replacement heart valves, other occlusion devices, and/or other medical implants, etc.

Embolic coils **104** may be typically introduced into a blood vessel by using a microcatheter (not explicitly shown) that extends from a proximal point outside the patient’s body to a distal point near the embolization site. An introducer sheath **106** containing the coil **104** can be used to carry and protect the coil **104** prior to insertion into the patient. Further, the introducer sheath **106** may be used to transfer the coil to the microcatheter and/or to assist in deploying the coil at a selected embolization site. The sheath **106** may be configured to protect the implant **104** and maintain the implant **104** in a deliverable orientation, until the implant **104** is deployed. As will be described in more detail herein, the locking element **108** may be configured to limit movement (e.g., axial and rotational) of the pusher wire **102** and implant **104** within the sheath **106** until the user is ready to advance the implant **104** out of the sheath **106**.

The sheath **106** may be a tubular member including a proximal end **112**, a distal end **114**, and an intermediate region **116** positioned therebetween. Some suitable but non-limiting materials for the sheath **106**, for example, polymer materials, composite materials, etc., are described below. The sheath **106** may define a lumen **110** extending from the proximal end **112** to the distal end **114**. The pusher wire **102** and implant **104** may be slidably disposed within the lumen **110** of the sheath **106** such that the pusher wire **102** and the implant **104** are radially inwards of the sheath **106**. The implant **104** may be disposed proximate the distal end **114** of the sheath **106**. The pusher wire **102** may be axially slidable between an interlocked position and a released position. The pusher wire **102** may be configured to be releasably attached to the implant **104**. The implant **104** may be configured to expand from a delivery configuration to a deployed configuration. The pusher wire **102** may generally be a solid wire or shaft, but may also be tubular in some embodiments. Some suitable but non-limiting materials for the pusher wire **102**, for example, metallic materials, polymer materials, composite materials, etc., are described below. As will be described in more detail herein, the pusher wire **102** may be releasably secured to the sheath **106** via the locking element **108** to limit axial and/or rotational movement of the pusher wire **102** within the sheath **106**.

The sheath **106** may have a first slot **118a** and a second slot **118b** (collectively, **118**) extending distally from the proximal end **112**. The first and second slots **118a**, **118b** may be positioned across from another or spaced approximately 180° about the circumference of the sheath **106**. While the sheath **106** is described as having two slots, the sheath **106** may include fewer than two or more than two slots, as desired. The slots **118** may extend less than an entire length of the sheath **106**. The slots **118** may remove material from the sheath **106** to create flexible arms or members **120a**, **120b** (collectively, **120**). While the sheath **106** is described as having two arms **120**, it should be understood the number of flexible arms may vary with the number of slots **118** and there can be fewer than two or more than two flexible arms, as desired. Further, the slots **118** may be uniformly or eccentrically distributed about a circumference of the sheath **106**. The length and/or size of the slots **118** (and/or the arms **120**) may vary to produce different degrees of wedging (between the locking element **108** and the pusher wire **102**) and locking capabilities. It is contemplated that a proximal end region **129** of the sheath **106** may be formed from polypropylene or a material similar thereto. The durometer of the proximal end region **129** may be manipulated to

produce a desired locking effect. In some cases, other portions of the sheath **106** may be formed from a same material as the proximal end region **129** while in other cases, other portions of the sheath **106** may be formed from a different material as the proximal end region **129**. For example, the sheath **106** may include a polyimide distal tip (although this is not required).

Referring additionally to FIG. 2, which illustrates a perspective view of the locking element **108**, the locking element **108** may be a tubular member having a proximal end **122**, a distal end **124**, and an intermediate region **126** positioned therebetween. The locking element **108** may have a generally constant or uniform outer diameter **130** over a proximal region **128** extending from the proximal end **122** towards and into the intermediate region **126**. The locking element **108** may include a distal region **132** including plurality of raised regions or circumferential ribs **136a**, **136b**, **136c**, **136d** (collectively, **136**). The circumferential ribs **136** may have an outer diameter **134** that is greater than the outer diameter **130** of the proximal region **128**. It is contemplated that the circumferential ribs **136** may increase the tactility of the locking element **108** and/or make the locking element **108** easier to grip or handle. The number of circumferential ribs **136** (e.g., fewer than four or greater than four), the size of the circumferential ribs **136** (e.g., increasing or decreasing the diameter **134**), the geometry of the circumferential ribs **136** (extending about less than an entire circumference, having a different shape, etc.), positioning of the circumferential ribs **136** along a length of the locking element **108**, and/or the spacing of the circumferential ribs **136** can be varied to enhance or diminish the tactility or gripability of the locking element **108**, as desired.

The locking element **108** may define a lumen **135** extending from the proximal end **122** to the distal end **124**. As will be discussed in more detail herein, a portion of the lumen **135** of the locking element **108** may be sized to slide freely over the sheath **106** while another portion of the lumen **135** may be sized to exert a radially inward compressing force on the sheath **106**, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath **106** may be radially inward of the locking element **108**.

The locking element **108** may be formed in a variety of different manners. For example, the locking element **108** may be injection molded, heat shrunk, 3-D printed, etc. The locking element **108** may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc.

FIG. 3 is a cross-sectional view of the locking element **108** taken at line 3-3 of FIG. 2. The locking element **108** may have a variable inner diameter. For example, in some embodiments, the locking element **108** may have an inner diameter which increases in a sloped manner over a first length **140** in the distal direction from a first inner diameter **137** adjacent the proximal end **122** to a second inner diameter **138**. In some embodiments, the first length **140** may be approximately the same as a length of the arms **120** of the sheath **106**, although this is not required. The second inner diameter **138** may be approximately constant or uniform over a second length **142** of the locking element **108**. In some cases, while not explicitly shown, the transition from the first inner diameter **137** to the second inner diameter **138** may be an abrupt or step-wise transition. The first length **140** and/or the second length **142** may vary, as desired. Further, the slope of the first length **140** may vary, as desired. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the tapered first length **140**

and/or the second length **142** may be changed to create the desired locking effect (described in more detail herein).

In some embodiments, a wall thickness of the locking element **108** may vary inversely to the inner diameter over the first length **140** while the outer diameter **130** remains constant to vary the inner diameter. For example, the locking element **108** may have a wall thickness **144** adjacent the proximal end **122**. The wall thickness may gradually decrease to a second wall thickness **146** at or near the second length **142**. In other embodiments, the wall thickness may remain constant. In such an instance, the inner diameter and the outer diameter of the locking element **108** may vary (e.g., may be sloped in a similar manner) along the first length **140**. For example, the outer diameter of the locking element **108** may increase from the proximal end **122** over the first length **140**, as the inner diameter increases.

The second, larger inner diameter **138** may be sized such that the locking element **108** can slide freely over the sheath **106**. The first, smaller inner diameter **137** may be sized to apply a compressive, or radially inward force on the outer surface of the sheath **106**, as will be described in more detail herein.

It is contemplated that the configuration of the locking element **108** may be adjusted to create the desired effect. For example, one or more of the inner diameters **137**, **138** may be made larger or smaller to accommodate different sizes of sheaths **106** and/or different outer diameters of the pusher wire **102**. Further, the first and/or second lengths **140**, **142** may be longer, shorter, less angled, more angled, etc. In another example, the outer diameter of the locking element **108** may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element **108** may include features in addition to, or in place of the circumferential ribs **136** to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath **106** and/or the locking element **108** may include visual indicia to guide the user in manipulation of the locking element **108**. In some cases, the overall length of the locking element **108** can be increased or decreased, as desired. It is further contemplated that the locking element **108** may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to, packaging.

FIG. 4 illustrates a perspective view of another illustrative locking element **200**. The locking element **200** may be similar in form and function to the illustrative locking mechanism **108** described herein and may be used with a medical device system, such as the medical device system **100** described herein. The locking element **200** may be a tubular member having a proximal end **202**, a distal end **204**, and an intermediate region **206** positioned therebetween. The locking element **200** may have a generally constant or uniform outer diameter **210** extending from the proximal end **202** to the distal end **204**, although this is not required. The locking element **200** may define a lumen **212** extending from the proximal end **202** to the distal end **204**. As will be discussed in more detail herein, a portion of the lumen **212** of the locking element **200** may be sized to slide freely over a sheath, such as the sheath **106** described herein, while another portion of the lumen **212** may be sized to exert a radially inward or compressive force on the sheath **106**, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath **106** may be radially inward of the locking element **200**.

The locking element **200** may be formed in a variety of different manners. For example, the locking element **200**

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may be injection molded, heat shrunk, 3-D printed, etc. The locking element 200 may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc.

FIG. 5 is a cross-sectional view of the locking element 200 taken at line 5-5 of FIG. 4. The locking element 200 may have a variable inner diameter. For example, in some embodiments, the locking element 200 may have an inner diameter which increases in a sloped manner over a first length 214 in the distal direction from a first inner diameter 216 adjacent the proximal end 202 to a second inner diameter 218. In some embodiments, the first length 214 may be approximately the same as a length of the arms 120 of the sheath 106, although this is not required. The second inner diameter 218 may be approximately constant or uniform over a second length 220 of the locking element 200. In some cases, while not explicitly shown, the transition from the first inner diameter 216 to the second inner diameter 218 may be an abrupt or step-wise transition. The first length 214 and/or the second length 220 may vary, as desired. Further, the slope of the first length 214 may vary, as desired. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the tapered first length 214 and/or the second length 220 may be changed to create the desired locking effect (described in more detail herein).

In some embodiments, a wall thickness of the locking element 200 may vary inversely to the inner diameter over the first length 214 while the outer diameter 210 remains constant to vary the inner diameter. For example, the locking element 200 may have a first wall thickness 222 adjacent the proximal end 202. The wall thickness may gradually decrease to a second wall thickness 224 at or near the second length 220. In other embodiments, the wall thickness may remain constant. In such an instance, the inner diameter and the outer diameter of the locking element 200 may vary (e.g., may be sloped in a similar manner) along the first length 214. For example, the outer diameter of the locking element 200 may increase from the proximal end 202 over the first length 214, as the inner diameter increases.

The second, larger inner diameter 218 may be sized such that the locking element 200 can slide freely over the sheath 106. The first, smaller inner diameter 216 may be sized to apply a compressive, or radially inward force on the outer surface of the sheath 106, as will be described in more detail herein.

It is contemplated that the configuration of the locking element 200 may be adjusted to create the desired effect. For example, one or more of the inner diameters 216, 218 may be made larger or smaller to accommodate different sizes of sheaths 106. Further, the first and/or second lengths 214, 220 may be longer, shorter, less angled, more angled, etc. In another example, the outer diameter of the locking element 200 may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element 200 may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath 106 and/or the locking element 200 may include visual indicia to guide the user in manipulation of the locking element 200. In some cases, the overall length of the locking element 200 can be increased or decreased, as desired. It is further contemplated that the locking element 200 may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.

FIG. 6 illustrates a perspective view of another illustrative locking element 300. The locking element 300 may be

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similar in form and function to the illustrative locking mechanism 108 described herein and may be used with a medical device system, such as the medical device system 100 described herein. The locking element 300 may be a tubular member having a proximal end 302, a distal end 304, and an intermediate region 305 positioned therebetween. The locking element 300 may define a lumen 312 extending from the proximal end 302 to the distal end 304. As will be discussed in more detail herein, a portion of the lumen 312 of the locking element 300 may be sized to slide freely over a sheath, such as the sheath 106 described herein, while another portion of the lumen 312 may be sized to exert a radially inward or compressive force on the sheath 106, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath 106 may be radially inward of the locking element 300. The locking element 300 may further include a slot 310 extending from the proximal end 302 to the distal end 304. The slot 310 may extend through a thickness of the side wall of the locking element 300 (e.g., from an outer surface to an inner surface thereof). The slot 310 may have a width 314 that is greater than a width of a pusher wire (e.g., such as the pusher wire 102 described herein) but less than a width of the sheath 106. This may allow the locking element 300 to be more easily removed from the medical device system 100 as will be described in more detail herein.

The locking element 300 may have a tapered or sloped outer diameter over a proximal region 328 extending from the proximal end 302 towards and into the intermediate region 305. The locking element 300 may include a distal region 330 including plurality of raised regions or circumferential ribs 332a, 332b, 332c, 332d (collectively, 332). The circumferential ribs 332 may have an outer diameter 334 that is greater than an outer diameter 324, 326 of the proximal region 328. The circumferential ribs 332 may extend about an entire perimeter of the locking element 300, which may be less than 360° due to the slot 310. It is contemplated that the circumferential ribs 332 may increase the tactility of the locking element 300 and/or make the locking element 300 easier to grip or handle. The number of circumferential ribs 332 (e.g., fewer than four or greater than four), the size of the circumferential ribs 332 (e.g., increasing or decreasing the diameter 334), the geometry of the circumferential ribs 332 (extending about less than an entire perimeter, having a different shape, etc.), and/or the spacing of the circumferential ribs 332 can be varied to enhance or diminish the tactility or gripability of the locking element 300, as desired.

The locking element 300 may be formed in a variety of different manners. For example, the locking element 300 may be injection molded, heat shrunk, 3-D printed, etc. The locking element 300 may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc.

FIG. 7 is a cross-sectional view of the locking element 300 taken at line 7-7 of FIG. 6. The locking element 300 may have a variable outer diameter and a variable inner diameter. For example, in some embodiments, the locking element 300 may have an inner diameter which increases in a sloped manner over a first length 316 in the distal direction from a first inner diameter 318 adjacent the proximal end 302 to a second inner diameter 320. In some embodiments, the first length 316 may be approximately the same as a length of the arms 120 of the sheath 106, although this is not required. The second inner diameter 320 may be approximately constant or uniform over a second length 322 of the locking element 300. In some cases, while not explicitly shown, the

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transition from the first inner diameter **318** to the second inner diameter **320** may be an abrupt or step-wise transition. The first length **316** and/or the second length **322** may vary, as desired. Further, the slope of the first length **316** may vary, as desired. The second, larger inner diameter **320** may be sized such that the locking element **300** can slide freely over the sheath **106**. The first, smaller inner diameter **318** may be sized to apply a compressive, or radially inward force on the outer surface of the sheath **106**, as will be described in more detail herein.

In some embodiments, a wall thickness **336** of the locking element **300** may remain generally uniform over the first and/or second lengths **316**, **322** of the locking element **300**. In such an instance, the inner diameter and the outer diameter of the locking element **300** may vary (e.g., may be sloped in a similar manner) along the first length **316**. For example, the outer diameter of the locking element **300** may increase from a first outer diameter **324** at the proximal end **302** to a second outer diameter **326** (greater than the first outer diameter **324**) over the first length **316**, as the inner diameter increases. In some embodiments, the wall thickness **336** may vary over the first and/or second lengths **316**, **322** of the locking element **300**. For example, when circumferential ribs **332** are provided, the wall thickness adjacent the ribs **332** may be greater than the wall thickness **336** of other portions of the locking element **300**. In other embodiments, the wall thickness **336** may vary to vary the diameter of the lumen **310**.

It is contemplated that the configuration of the locking element **300** may be adjusted to create the desired effect. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the tapered first length **316** and/or the second length **322** may be changed to create the desired locking effect (described in more detail herein). For example, one or more of the inner diameters **318**, **320** may be made larger or smaller to accommodate different sizes of sheaths **106**. Further, the first and/or second lengths **316**, **322** may be longer, shorter, less angled, more angled, etc. In another example, the outer diameter of the locking element **300** may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element **300** may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath **106** and/or the locking element **300** may include visual indicia to guide the user in manipulation of the locking element **300**. In some cases, the overall length of the locking element **300** can be increased or decreased, as desired. It is further contemplated that the locking element **300** may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.

FIG. **8** is a perspective view of the illustrative medical device system **100** in an unlocked or first assembled configuration and FIG. **9** is a partial cross-sectional view of the illustrative medical device system **100**, taken at line **9-9** of FIG. **8**. While the medical device system **100** is described with respect to a particular locking element **108**, it should be understood that any of the locking elements described herein, including but not limited to, the locking elements **200**, **300**, **408**, **508**, **600**, **708**, **800**, may be substituted for the locking element **108**. From the partially unassembled configuration illustrated in FIG. **1**, to position the locking element **108** such that it can be used to secure the pusher wire **102**, the locking element **108** is distally advanced over a proximal end region **129** of the sheath **106** from the proximal end **112**. An entirety of the locking element **108**

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may not be disposed over sheath **106** in the first assembled configuration. For example, the first length **140** of the locking element **108** having the reduced inner diameter may be only partially disposed over the sheath **106** or not over the sheath **106** at all. In this configuration, the pusher wire **102** is free to slide axially (e.g., proximally and distally) and/or rotate within the lumen **110** of the sheath **106**. As described above, the inner diameter of the sheath **106** may remain constant, or substantially constant from the proximal end **112** to the distal end **114**.

To lock the pusher wire **102** relative to the sheath **106**, the locking element **108** is advanced further in the distal direction until the first length **140** of the locking element **108** is disposed over the arms **120** of the sheath **106**. FIG. **10** is a perspective view of the illustrative medical device system **100** in a locked or second assembled configuration and FIG. **11** is a partial cross-sectional view of the illustrative medical device system **100**, taken at line **11-11** of FIG. **10**. As described above, the distal portion **132** and a portion of the proximal portion **128** of locking element **108** may slide freely over the proximal end region **129** of the sheath. When proximal portion **128** having the reduced diameter of the locking element **108** reaches the outer sheath **106**, the arms **120** of the sheath **106** may be depressed onto an outer surface of the pusher wire **102**. For example, the locking element **108** may freely slide over the outer sheath **106** until the inner diameter of the locking element **108** is approximately equal to or less than an outer diameter of the sheath **106**. It is contemplated that the wall thickness of the outer sheath **106** and the outer diameter of the pusher wire **102** may collectively affect how much the sheath arms **120** collapse before locking on the pusher wire **102**. Once the inner diameter of the locking element **108** begins to frictionally engage the outer surface of the sheath **106** more force may be required to continue distal movement of the locking element **108**. As the locking element **108** is distally advanced further onto the sheath **106**, the tapered region (e.g., the first length **140**) acts to deflect the arms **120** of the sheath **106** inward and onto the wire **102**. When advanced far enough, the locking element **108** wedges in place, creating the desired locking effect between the sheath **106** and the pusher wire **102**. For example, the tapered portion of the lumen **135** of the locking element **108** may act as a wedge to pinch or exert a radially inward biasing force on an outer surface of the arms **120** of the sheath **106** such that the outer diameter of the arms **120** is decreased and the inner diameter of the arms is biased radially inward, as shown in FIG. **11**. As the inner diameter of the arms **120** is reduced, the inner surface of the arms **120** contacts and frictionally engages an outer surface of the pusher wire **102**.

In the locked configuration illustrated in FIGS. **10** and **11**, the frictional engagement between the inner surface of the sheath **106** and the outer surface of the pusher wire **102** may preclude or inhibit axial (e.g., proximally and distally) and/or rotational movement of the pusher wire **102** within the lumen **110** of the sheath **106**. In some embodiments, the arms **120** are configured to promote inward deflection thereof. For example, the cut pattern of the slots **118** may be applied to promote collapse of the arms **120** and to create a localized friction point between an inner surface of the sheath **106** (e.g., the inner surface of the arms **120**) and the pusher wire **102**.

It is contemplated that the gradual decrease in the inner diameter of the locking element **108** may facilitate positioning of the locking element **108** over the sheath **106**. For example, the inner diameter **138** of the locking element **108** at the distal portion **132** (and/or portions of the proximal

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portion **128** may be large enough to freely pass over the outer diameter of the sheath **106**. As the inner diameter of the locking element tapers (e.g., in the proximal direction towards the proximal end **122**) to a dimension smaller than the outer diameter of the sheath **106**, and is distally advanced far enough over the sheath **106**, it promotes collapse of the arms **120**. This creates a friction lock between the sheath **106** and pusher wire **102** when the pusher wire **102** is positioned through the sheath **106**.

When the user is ready to advance the implant **104**, the locking element **108** may be removed proximally from the proximal region **129** of the sheath **106**. This may allow the inner diameter of the sheath **106** adjacent to the arms **120** to expand to the original configuration thus removing the friction lock and allowing the pusher wire **102** to move freely. Thus, as the sheath **106** is removed from the pusher wire **102** and/or the pusher wire **102** is distally advanced, the sheath **106** does not engage or hang up on any portion of the pusher wire **102**.

It is contemplated that when the locking element **108** is provided with a longitudinally extending slot, such as the slot **310** illustrated in the locking element **300**, the locking element **108** may be free to drop off of the pusher wire **102** (e.g., via the slot) without having to be proximally retracted over an entirety of the length of the pusher wire **102** proximal to the sheath **106**. Similarly, during loading of the locking element **108** at the time of manufacture, the locking element **108** can be placed over the pusher wire **102** rather than threading the locking element **108** over the pusher wire **102**. As described above with respect to FIGS. **6** and **7**, if so provided a longitudinal slot may be sized such that the pusher wire **102** can fit through the slot, but not the sheath **106**. This may ensure the locking element **108** does not fall off the sheath **106** in the locked position (e.g., FIGS. **10** and **11**) but can be easily removed from the pusher wire **102** once the locking element is in the unlocked configuration.

It is further contemplated that that locking element **108** may be formed from a material that is more rigid than or stiffer than the outer sheath **106**. For example, the material for the locking element **108** may be selected such that the locking element **108** does not deflect when the tapered portion of the lumen **135** of the locking element **108** is advanced over the sheath **106** but rather forces the sheath **106** to deflect inward. In some embodiments, the locking element **108** may be formed from a bright (or other) color easily noticeable by the user.

FIG. **12** is a perspective view of another illustrative medical device system **400** in a partially unassembled configuration. FIG. **13** is a cross-sectional view of the illustrative medical device system **400** taken at line **13-13** of FIG. **12**. The medical device system **400** may include a pusher wire **402**, an implant (not explicitly shown), such as, but not limited to, an embolic coil, an introducer sheath **406**, and a locking element **408**. For simplicity, the implant is described as an embolic coil, but other suitable medical devices transported, delivered, used, released, etc. in a similar manner are also contemplated, including but not limited to, vascular occlusion devices coils, stents, embolic filters, replacement heart valves, other occlusion devices, and/or other medical implants, etc.

Embolic coils may be typically introduced into a blood vessel by using a microcatheter (not explicitly shown) that extends from a proximal point outside the patient's body to a distal point near the embolization site. An introducer sheath **406** containing the coil can be used to carry and protect the coil prior to insertion into the patient. Further, the introducer sheath **406** may be used to transfer the coil to the

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microcatheter and/or to assist in deploying the coil at a selected embolization site. The sheath **406** may be configured to protect the implant and maintain the implant in a deliverable orientation, until the implant is deployed. As will be described in more detail herein, the locking element **408** may be configured to limit movement (e.g., axial and rotational) of the pusher wire **402** and implant within the sheath **406** until the user is ready to advance the implant out of the sheath **406**.

The sheath **406** may be a tubular member including a proximal end **412**, a distal end **414**, and an intermediate region **415** positioned therebetween. Some suitable but non-limiting materials for the sheath **406**, for example, polymer materials, composite materials, etc., are described below. The sheath **406** may define a lumen **410** extending from the proximal end **412** to the distal end **414**. The pusher wire **402** and implant may be slidably disposed within the lumen **410** of the sheath **406** such that the pusher wire **402** and the implant are radially inwards of the sheath **406**. The implant may be disposed proximate the distal end **414** of the sheath **406**. The pusher wire **402** may be axially slidable between an interlocked position and a released position. The pusher wire **402** may be configured to be releasably attached to the implant. The implant may be configured to expand from a delivery configuration to a deployed configuration. The pusher wire **402** may generally be a solid wire or shaft, but may also be tubular in some embodiments. Some suitable but non-limiting materials for the pusher wire **402**, for example metallic materials, polymer materials, composite materials, etc., are described below. As will be described in more detail herein, the pusher wire **402** may be releasably secured to the sheath **406** via the locking element **408** to limit axial and/or rotational movement of the pusher wire **402** within the sheath **406**.

The sheath **406** may have a longitudinally extending slot **416** extending distally from the proximal end **412**. The slot **416** may terminate at a "C" cut **418** extending about a circumference of the sheath **416**. The cut **418** may be positioned in a plane that is generally perpendicular to the longitudinal axis of the sheath **406**. In some cases, the cut **418** may extend less than 360°, less than 270°, less than 180°, less than 135°, less than 90° about the circumference of the sheath **406**, as desired. It is contemplated that the radial length of the cut **418** may be varied to produce a desired locking effect. Similarly, the length and width of the slot **416** may be varied to produce a desired locking effect. The slot **416** and the cut **418** may allow the free edges or flaps **420a**, **420b** (collectively, **420**) of the sheath **406** to fold over or collapse when a radially inward force is applied to an outer surface thereof to create a frictional lock, as will be described in more detail here.

It is contemplated that a proximal end region **422** of the sheath **406** may be formed from polypropylene or a material similar thereto. The durometer of the proximal end region **422** may be manipulated to produce a desired locking effect. In some cases, other portions of the sheath **406** may be formed from a same material as the proximal end region **422** while in other cases, other portions of the sheath **406** may be formed from a different material as the proximal end region **422**. For example, the sheath **406** may include a polyimide distal tip (although this is not required).

Referring additionally to FIG. **14**, which illustrates a perspective view of the locking element **408**, the locking element **408** may be a tubular member having a proximal end **424**, a distal end **426**, and an intermediate region **428** positioned therebetween. The locking element **408** may have a generally constant or uniform outer diameter **430** over the

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intermediate region 428. The outer diameter may increase from the intermediate region 428 in both the proximal and distal directions to create a flared proximal portion 432 and a flared distal portion 434. The flared proximal portion 432 may increase in diameter towards the proximal end 424 to a second outer diameter 436 greater than the first outer diameter 430 adjacent to the intermediate region 428. Similarly, the flared distal portion 434 may increase in diameter towards the distal end 426 to the second outer diameter 436 greater than the first outer diameter 430 adjacent to the intermediate region 428.

The locking element 408 may define a lumen 438 extending from the proximal end 424 to the distal end 426. As will be discussed in more detail herein, a portion of the lumen 438 of the locking element 408 may be sized to slide freely over the sheath 406 while another portion of the lumen 438 may be sized to exert a radially inward compressing force on the sheath 406, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath 406 may be radially inward of the locking element 408.

The locking element 408 may be formed in a variety of different manners. For example, the locking element 408 may be injection molded, heat shrunk, 3-D printed, etc. The locking element 408 may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc.

FIG. 15 is a cross-sectional view of the locking element 408 taken at line 15-15 of FIG. 14. The locking element 408 may have a variable inner diameter. In some cases, the lumen 438 may have a generally hourglass shape. For example, in some embodiments, the locking element 408 may have an inner diameter which has a first inner diameter 440 adjacent to the proximal and distal ends 424, 426 thereof. The first inner diameter 440 may taper or slope towards a second generally constant inner diameter 442. It is contemplated that the sloped inner diameter may generally correspond to the flared proximal and distal portions 432, 434 while the generally constant inner diameter 442 may generally correspond to the intermediate region 428. However, this is not required. However, in such an arrangement, the locking element 408 may have a generally uniform wall thickness from the proximal end 424 to the distal end 426 thereof. In other embodiments the inner diameter of the locking element 408 may be varied by varying a wall thickness of the locking element 408 along a length thereof. Further, in some cases, while not explicitly shown, the transition from the first inner diameter 440 to the second inner diameter 442 may be an abrupt or step-wise transition. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the flared proximal portion 432, the flared distal portion 434 and/or the intermediate region 428 may be changed to create the desired locking effect (described in more detail herein).

The first, larger inner diameter 440 may be sized such that the locking element 408 can slide freely over the sheath 406. The first, smaller inner diameter 442 may be sized to apply a compressive, or radially inward force on the outer surface of the sheath 406, as will be described in more detail herein.

It is contemplated that the configuration of the locking element 408 may be adjusted to create the desired effect. For example, one or more of the inner diameters 440, 442 may be made larger or smaller to accommodate different sizes of sheaths 406. Further, the flared proximal portion 432, the flared distal portion 434 and/or the intermediate region 428 may be longer, shorter, less angled, more angled, etc. In another example, the outer diameter of the locking element 408 may be increased or decreased to facilitate handling. It

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is further contemplated that the outer surface of the locking element 408 may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath 406 and/or the locking element 408 may include visual indicia to guide the user in manipulation of the locking element 408. In some cases, the overall length of the locking element 408 can be increased or decreased, as desired. It is further contemplated that the locking element 408 may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.

FIG. 16 is a perspective view of the illustrative medical device system 400 in a locked configuration and FIG. 17A is a partial cross-sectional view of the illustrative medical device system 400, taken at line 17A-17A of FIG. 16. While the medical device system 400 is described with respect to a particular locking element 408, it should be understood that any of the locking elements described herein, including but not limited to the locking elements 108, 200, 300, 508, 600, 708, 800, may be substituted for the locking element 408. From the partially unassembled configuration illustrated in FIG. 12, to position the locking element 408 such that it can be used to secure the pusher wire 402, the locking element 408 is distally advanced over a proximal end region 422 of the sheath 406 from the proximal end 412.

To lock the pusher wire 402 relative to the sheath 406, the locking element 408 is advanced further in the distal direction until the intermediate region 428 of the locking element 408 is disposed over the flaps 420 of the sheath 406. The flared distal portion 432 of locking element 408 may slide freely over the proximal end region 422 of the sheath. When the intermediate region 428 having the reduced diameter 442 of the locking element 408 reaches the outer sheath 406, the flaps 420 of the sheath 406 may fold over and be depressed onto an outer surface of the pusher wire 402. This is more clearly illustrated in FIG. 17B, which is a partial cross-section view of the illustrative medical device system 400, taken at line 17B-17B of FIG. 17A. For example, the locking element 408 may freely slide over the outer sheath 406 until the inner diameter of the locking element 408 is approximately equal to or less than an outer diameter of the sheath 406. Once the inner diameter of the locking element 408 begins to frictionally engage the outer surface of the sheath 406 more force may be required to continue distal movement of the locking element 408. As the locking element 408 is distally advanced further onto the sheath 406, the intermediate region 428 acts to deflect the flaps 420 of the sheath 406 inward and onto the wire 402. In some cases, the tapered lumen of the flared distal portion 434 may gradually deflect the flaps 420 inward. When advanced far enough, the locking element 408 wedges in place, creating the desired locking effect between the sheath 406 and the pusher wire 402. For example, the reduced diameter portion (e.g., having the second smaller inner diameter 442) of the lumen 438 of the locking element 408 may act as a wedge to pinch or exert a radially inward biasing force on an outer surface of the flaps 420 of the sheath 406 such that the outer diameter of the flaps 420 is decreased and the inner diameter of the arms is biased radially inward, as shown in FIG. 17B. As the inner diameter of the flaps 420 is reduced, the inner surface of the flaps 420 contacts and frictionally engages an outer surface of the pusher wire 402.

In the locked configuration illustrated in FIGS. 16, 17A, and 17B, the frictional engagement between the inner surface of the sheath 406 and the outer surface of the pusher wire 402 may preclude or inhibit axial (e.g., proximal and

distal) and/or rotational movement of the pusher wire **402** within the lumen **410** of the sheath **406**. In some embodiments, the flaps **420** are configured to promote inward deflection thereof. For example, the cut pattern of the slot **416** and/or slit **418** may be applied to promote collapse of the flaps **420** and to create a localized friction point between an inner surface of the sheath **406** (e.g., the inner surface of the flaps **420**) and the pusher wire **402**.

It is contemplated that the gradual decrease in the inner diameter of the locking element **408** may facilitate positioning of the locking element **408** over the sheath **406**. For example, the inner diameter **440** of the locking element **408** at the distal portion **434** (and/or portions of the proximal portion **432** may be large enough to freely pass over the outer diameter of the sheath **406**. As the inner diameter of the locking element **408** tapers (e.g., in the proximal direction towards the proximal end **424**) to a dimension smaller than the outer diameter of the sheath **406**, and is distally advanced far enough over the sheath **406**, it promotes collapse of the flaps **420**. This creates a friction lock between the sheath **406** and pusher wire **402** when the pusher wire **402** is positioned through the sheath **406**.

When the user is ready to advance the implant, the locking element **408** may be removed proximally from the proximal region **422** of the sheath **406**. This may allow the inner diameter of the sheath **406** adjacent to the flaps **420** to expand to the original configuration thus removing the friction lock and allowing the pusher wire **402** to move freely. Thus, as the sheath **406** is removed from the pusher wire **402**, the sheath **406** does not engage or hang up on any portion of the pusher wire **402**.

It is contemplated that when the locking element **408** is provided with a longitudinally extending slot, such as the slot **310** illustrated in the locking element **300**, the locking element **408** may be free to drop off of the pusher wire **402** (e.g., via the slot) without having to be proximally retracted over an entirety of the length of the pusher wire **402** proximal to the sheath **406**. Similarly, during loading of the locking element **408** at the time of manufacture, the locking element **408** can be placed over the pusher wire **402** rather than threading the locking element **408** over the pusher wire **402**. As described above with respect to FIGS. **6** and **7**, if so provided, a longitudinal slot may be sized such that the pusher wire **402** can fit through the slot, but not the sheath **406**. This may ensure the locking element **408** does not fall off the sheath **406** in the locked position (e.g., FIGS. **16**, **17A**, and **17B**) but can be easily removed from the pusher wire **402** once the locking element is in the unlocked configuration.

It is further contemplated that that locking element **408** may be formed from a material that is more rigid than or stiffer than the outer sheath **406**. For example, the material for the locking element **408** may be selected such that the locking element **408** does not deflect when the tapered portion of the lumen **438** of the locking element **408** is advanced over the sheath **406** but rather forces the sheath **406** to deflect inward. In some embodiments, the locking element **408** may be formed from a bright (or other) color easily noticeable by the user.

FIG. **18** is a perspective view of an example medical device system **500** in an unlocked assembled configuration. FIG. **19** is a partial cross-sectional view of the medical device system **500** taken at line **19-19** of FIG. **18**. The medical device system **500** may include a pusher wire **502**, an implant (not explicitly shown), such as, but not limited to, an embolic coil, an introducer sheath **506**, and a locking element **508**. For simplicity, the implant is described as an

embolic coil, but other suitable medical devices transported, delivered, used, released, etc. in a similar manner are also contemplated, including but not limited to, vascular occlusion devices coils, stents, embolic filters, replacement heart valves, other occlusion devices, and/or other medical implants, etc.

Embolic coils may be typically introduced into a blood vessel by using a microcatheter (not explicitly shown) that extends from a proximal point outside the patient's body to a distal point near the embolization site. An introducer sheath **506** containing the coil can be used to carry and protect the coil prior to insertion into the patient. Further, the introducer sheath **506** may be used to transfer the coil to the microcatheter and/or to assist in deploying the coil at a selected embolization site. The sheath **506** may be configured to protect the implant and maintain the implant in a deliverable orientation, until the implant is deployed. As will be described in more detail herein, the locking element **508** may be configured to limit movement (e.g., axial and rotational) of the pusher wire **502** and implant within the sheath **506** until the user is ready to advance the implant out of the sheath **506**.

The sheath **506** may be a tubular member including a proximal end **512** (e.g., FIG. **19**), a distal end **514**, and an intermediate region **516** positioned therebetween. Some suitable but non-limiting materials for the sheath **506**, for example, polymer materials, composite materials, etc., are described below. The sheath **506** may define a lumen **510** extending from the proximal end **512** to the distal end **514**. The pusher wire **502** and implant may be slidably disposed within the lumen **510** of the sheath **506** such that the pusher wire **502** and the implant are radially inwards of the sheath **506**. The implant may be disposed proximate the distal end **514** of the sheath **506**. The pusher wire **502** may be axially slidable between an interlocked position and a released position. The pusher wire **502** may be configured to be releasably attached to the implant. The implant may be configured to expand from a delivery configuration to a deployed configuration. The pusher wire **502** may generally be a solid wire or shaft, but may also be tubular in some embodiments. Some suitable but non-limiting materials for the pusher wire **502**, for example metallic materials, polymer materials, composite materials, etc., are described below. As will be described in more detail herein, the pusher wire **502** may be releasably secured to the sheath **506** via the locking element **508** to limit axial and/or rotationally movement of the pusher wire **502** within the sheath **506**.

It is contemplated that a proximal end region **520** of the sheath **506** may be formed from polypropylene or a material similar thereto. The durometer of the proximal end region **520** may be manipulated to produce a desired locking effect. In some cases, other portions of the sheath **506** may be formed from a same material as the proximal end region **520** while in other cases, other portions of the sheath **506** may be formed from a different material as the proximal end region **520**. For example, the sheath **506** may include a polyimide distal tip (although this is not required).

Referring additionally to FIG. **19**, the sheath **506** may include an angled protrusion **518** adjacent the proximal end **512** and extending radially outward from an outer surface of the sheath **506**. The angled protrusion **518** may cooperate with a corresponding protrusion **528** on the locking element **508** to allow the locking element **508** to be distally advanced over the sheath **506** while providing a mechanical stop that limits actuation of the locking element **508** in the proximal direction. In some instances, the angled protrusion **518** may be a single component extending about an entire circumfer-

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ence (e.g., 360°) of the sheath 506. In other cases, the angled protrusion 518 may be formed from a plurality of individual protrusions which are spaced (uniformly or eccentrically) about the circumference of the sheath 506. The angled protrusion 518 may have an outer diameter which decreases in the proximal direction (e.g., towards the proximal end 512). The distal end of the angled protrusion 518 may be generally planar and have an outer diameter that is greater than an inner diameter of the proximal end of the protrusion 528 of the locking element 508, as will be described in more detail herein. The distal end of the angled protrusion 518 may extend generally orthogonal to a longitudinal axis of the sheath 506 while the angled surface extends at an angle that is non-orthogonal and non-parallel to the longitudinal axis of the sheath 506.

The locking element 508 may be a tubular member having a proximal end 522, a distal end 524, and an intermediate region 526 positioned therebetween. The locking element 508 may have a tapered or sloped outer diameter over a proximal region 536 that increases from the proximal end 522 towards and into the intermediate region 526. The locking element 508 may include a distal region 538 having a generally uniform or constant outer diameter. The locking element 508 may define a lumen 530 extending from the proximal end 522 to the distal end 524. As will be discussed in more detail herein, a portion of the lumen 530 of the locking element 508 may be sized to slide freely over the sheath 506 while another portion of the lumen 530 may be sized to exert a radially inward compressing force on the sheath 506, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath 506 may be radially inward of the locking element 508.

The locking element 508 may be formed in a variety of different manners. For example, the locking element 508 may be injection molded, heat shrunk, 3-D printed, etc. The locking element 508 may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc.

The locking element 508 may have a variable inner diameter. For example, in some embodiments, the locking element 508 may have an inner diameter which increases in a sloped manner over the proximal region 536 in the distal direction from a first, smaller, inner diameter adjacent the proximal end 522 to a second, larger, inner diameter. The second inner diameter 138 may be approximately constant or uniform over a distal region 538 of the locking element 508. In some cases, while not explicitly shown, the transition from the first inner diameter to the second inner diameter 138 may be an abrupt or step-wise transition. The proximal region 536 and/or the distal region 538 may vary, as desired. Further, the slope of the proximal region 536 may vary, as desired. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the tapered proximal region 536 and/or the distal region 538 may be changed to create the desired locking effect (described in more detail herein).

In some embodiments, a wall thickness of the locking element 508 may remain generally uniform over the proximal and/or distal regions 536, 538 of the locking element 508. In such an instance, the inner diameter and the outer diameter of the locking element 508 may vary (e.g., may be sloped in a similar manner) along the proximal region 536. For example, the outer diameter of the locking element 508 may increase from a first outer diameter at the proximal end 522 to a second outer diameter (greater than the first outer diameter) over the proximal region 536, as the inner diameter increases. In some embodiments, the wall thickness may vary over the proximal and/or distal regions 536, 538 of the

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locking element 508. The second, larger, inner diameter may be sized such that the locking element 508 can slide freely over the sheath 506. The first, smaller, inner diameter may be sized to apply a compressive, or radially inward force on the outer surface of the sheath 506, as will be described in more detail herein.

The locking element 508 may include an angled protrusion 528 adjacent the distal end 524 and extending radially inward from an inner surface of the locking element 508. The angled protrusion 528 may cooperate with the protrusion 518 on the sheath 506 to allow the locking element 508 to be distally advanced over the sheath 506 while providing a mechanical stop that limits actuation of the locking element 508 in the proximal direction. In some instances, the angled protrusion 528 may be a single component extending about an entire inner surface of the lumen 530 (e.g., 360°) of the locking element 508. In other cases, the angled protrusion 528 may be formed from a plurality of individual protrusions which are spaced (uniformly or eccentrically) about the inner surface of the lumen 530. The angled protrusion 528 may have an inner diameter which decreases in the distal direction (e.g., towards the distal end 524). The proximal end of the angled protrusion 528 may be generally planar and have an inner diameter that is less than an outer diameter of the distal end of the protrusion 518 of the sheath 506. The proximal end of the angled protrusion 528 may extend generally orthogonal to a longitudinal axis of the locking element 508 while the angled surface extends at an angle that is non-orthogonal and non-parallel to the longitudinal axis of the locking element 508.

The sloped surfaces of the angled protrusions 518, 528 may be structured such that the angled protrusion 518 on the sheath 506 essentially functions as a ramp for the angled protrusion 528 on the locking element 508. This may allow the locking element 508 to be loaded onto the proximal end region 520 of the sheath 506 by advancing the distal end 512 of the sheath 506 through the distal end 524 of the locking element. Once loaded onto the sheath 506, proximal retraction of the locking element 508 will be limited. For example, the generally planar proximal end of the angled protrusion 528 on the locking element 508 will abut the generally planar distal end of the angled protrusion 518 on the sheath 506 creating a mechanical stop thus limiting the axial movement of the locking element 508. This may allow the locking element 508 to be moved between the locked (see, for example, FIG. 20) and unlocked configuration over a shorter distance and without fully removing it from the pusher wire 502.

It is contemplated that the configuration of the locking element 508 may be adjusted to create the desired effect. For example, one or more of the inner diameters may be made larger or smaller to accommodate different sizes of sheaths 506. Further, the proximal and/or distal regions 536, 538 may be longer, shorter, less angled, more angled, etc. In another example, the outer diameter of the locking element 508 may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element 508 may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath 506 and/or the locking element 508 may include visual indicia to guide the user in manipulation of the locking element 508. In some cases, the overall length of the locking element 508 can be increased or decreased, as desired. It is further contemplated that the locking element 508 may be large enough to make it easy to handle but not

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so large as to be incompatible with other components, such as, but not limited to packaging.

While the medical device system 500 is described with respect to a particular locking element 508, it should be understood that any of the locking elements described herein may be substituted for the locking element 508. To position the locking element 508 such that it can be used to secure the pusher wire 502, the locking element 508 is distally advanced over the proximal end region 520 of the sheath 506 from the proximal end 512. It should be understood that the varying lengths of the locking element 508 may be positioned over the sheath 506 in an unlocked configuration. However, an entirety of the locking element 508 may not be disposed over sheath 506 in the unlocked configuration. For example, the proximal region 536 of the locking element 508 having the reduced inner diameter may be only partially disposed over the sheath 506 or not over the sheath 506 at all. In this configuration, the pusher wire 502 is free to slide axially (e.g., proximally and distally) and/or rotate within the lumen 510 of the sheath 506. As described above, the inner diameter of the sheath 506 may remain constant, or substantially constant from the proximal end 512 to the distal end 514.

To lock the pusher wire 502 relative to the sheath 506, the locking element 508 is advanced further in the distal direction until the proximal region 536 of the locking element 508 is disposed over the proximal end region 520 of the sheath 506. FIG. 20 is a partial cross-sectional view of the medical device system 500 in a locked configuration. As described above, the distal portion 538 and a portion of the proximal portion 536 of locking element 508 may slide freely over the proximal end region 520 of the sheath. When proximal portion 536 having the reduced diameter of the locking element 508 reaches the outer sheath 506, the sheath 506 may be depressed onto an outer surface of the pusher wire 502. For example, the locking element 508 may freely slide over the outer sheath 506 until the inner diameter of the locking element 508 is approximately equal to or less than an outer diameter of the sheath 506. Once the inner diameter of the locking element 508 begins to frictionally engage the outer surface of the sheath 506 more force may be required to continue distal movement of the locking element 508. As the locking element 508 is distally advanced further onto the sheath 506, the tapered region (e.g., the proximal region 536) acts to deflect the sheath 506 inward and onto the wire 502. When advanced far enough, the locking element 508 wedges in place, creating the desired locking effect between the sheath 506 and the pusher wire 502. For example, the tapered portion of the lumen 530 of the locking element 508 may act as a wedge to pinch or exert a radially inward biasing force on an outer surface of the sheath 506 such that the outer diameter of sheath 506 is decreased and the inner diameter of the sheath 506 is biased radially inward, as shown in FIG. 20. As the inner diameter of the sheath 506 is reduced, the inner surface of the sheath 506 contacts and frictionally engages an outer surface of the pusher wire 502.

In the locked configuration illustrated in FIG. 20, the frictional engagement between the inner surface of the sheath 506 and the outer surface of the pusher wire 502 may preclude or inhibit axial (e.g., proximally and distally) and/or rotational movement of the pusher wire 502 within the lumen 510 of the sheath 506. In some embodiments, the sheath 506 may be configured to promote inward deflection thereof. For example, the sheath 506 may include cuts, slots, have a reduced thickness, be more flexible, etc. to promote

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collapse of the sheath 506 and to create a localized friction point between an inner surface of the sheath 506 and the pusher wire 502.

It is contemplated that the gradual decrease in the inner diameter of the locking element 508 may facilitate positioning of the locking element 508 over the sheath 506. For example, the inner diameter of the locking element 508 at the distal portion 538 (and/or portions of the proximal portion 536) may be large enough to freely pass over the outer diameter of the sheath 506. As the inner diameter of the locking element tapers (e.g., in the proximal direction towards the proximal end 522) to a dimension smaller than the outer diameter of the sheath 506, and is distally advanced far enough over the sheath 506, it promotes collapse of the sheath 506. This creates a friction lock between the sheath 506 and pusher wire 502 when the pusher wire 502 is positioned through the sheath 506.

When the user is ready to advance the implant, the locking element 508 may be proximally displaced (but not entirely removed) from the proximal region 520 of the sheath 506. This may allow the inner diameter of the sheath 506 to expand to the original configuration thus removing the friction lock and allowing the pusher wire 502 to move freely. Thus, as the sheath 506 is removed from the pusher wire 502, the sheath 506 does not engage or hang up on any portion of the pusher wire 502.

It is contemplated that the locking element 508 may be formed from a material that is more rigid than or stiffer than the outer sheath 506. For example, the material for the locking element 508 may be selected such that the locking element 508 does not deflect when the tapered portion of the lumen 530 of the locking element 508 is advanced over the sheath 506 but rather forces the sheath 506 to deflect inward. In some embodiments, the locking element 508 may be formed from a bright (or other) color easily noticeable by the user.

FIG. 21 illustrates a perspective view of another illustrative locking element 600. The locking element 600 may be similar in form and function to the illustrative locking mechanism 108 described herein and may be used with a medical device system, such as the medical device system 100 described herein. The locking element 600 may be a tubular member having a proximal end 602, a distal end 604, and an intermediate region 605 positioned therebetween. The locking element 600 may define a lumen 612 extending from the proximal end 602 to the distal end 604. A portion of the lumen 612 of the locking element 600 may be sized to slide freely over a sheath, such as the sheath 106 described herein, while another portion of the lumen 612 may be sized to exert a radially inward or compressive force on the sheath 106, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath 106 may be radially inward of the locking element 600.

The locking element 600 may have a tapered or sloped outer diameter over a proximal region 628 extending from the proximal end 602 towards and into the intermediate region 605. Similarly, the locking element 600 may have a tapered or sloped outer diameter over a distal region 630 extending from the distal end 604 towards and into the intermediate region 605. However, in some cases, one or both the proximal region 628 and the distal region 630 may have a generally constant outer diameter or the diameter may increase towards the intermediate region 605 as desired. The intermediate region 605 may include plurality of raised regions or circumferential ribs 632a, 632b, 632c (collectively, 632). The circumferential ribs 632 may have an outer diameter 634 that is greater than an outer diameter 624 of the

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intermediate region 605 between the ribs 632. The circumferential ribs 632 may extend about an entire perimeter of the locking element 600 or less than an entire perimeter, as desired. It is contemplated that the circumferential ribs 632 may increase the tactility of the locking element 600 and/or make the locking element 600 easier to grip or handle. The number of circumferential ribs 632 (e.g., fewer than three or greater than three), the size of the circumferential ribs 632 (e.g., increasing or decreasing the diameter 634), the geometry of the circumferential ribs 632 (extending about less than an entire perimeter, having a different shape, etc.), and/or the spacing of the circumferential ribs 632 can be varied to enhance or diminish the tactility or gripability of the locking element 600, as desired.

The locking element 600 may be formed in a variety of different manners. For example, the locking element 600 may be injection molded, heat shrunk, 3-D printed, etc. The locking element 600 may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc.

FIG. 22 is a cross-sectional view of the locking element 600 taken at line 22-22 of FIG. 21. The locking element 600 may have a variable outer diameter and a variable inner diameter. For example, in some embodiments, the locking element 600 may have an inner diameter which increases in a sloped manner in the distal direction from a first inner diameter 618 adjacent the proximal end 602 to a second inner diameter 620 adjacent the distal end 604. Further, the slope of the inner surface may vary, as desired. The second, larger inner diameter 620 may be sized such that the distal region 630 of the locking element 600 can slide freely over the sheath 106. The first, smaller inner diameter 618 may be sized such that the proximal region 628 is configured to apply a compressive, or radially inward force on the outer surface of the sheath 106.

In some embodiments, a wall thickness 636 of the locking element 600 may vary over a length of the locking element 600. In some cases, the wall thickness 636 is not directly correlated to the inner diameter of the locking element 600. In other cases, the wall thickness 636 may reduce in size as the inner diameter increases. These are just some examples. The outer diameter of the locking element 600 may decrease from a first outer diameter 624 at the proximal end 602 to a second outer diameter 626 (less than the first outer diameter 624) adjacent to the intermediate region 605. The outer diameter of the locking element 600 may decrease from a third outer diameter 636 at the proximal end 602 to a fourth outer diameter 632 (less than the third outer diameter 636) adjacent to the intermediate region 605. In some embodiments the first outer diameter 622 and the third outer diameter 636 may be approximately equal, although this is not required. Similarly, the second outer diameter 626 and the fourth outer diameter 632 may be approximately equal, although this is not required.

It is contemplated that the configuration of the locking element 600 may be adjusted to create the desired effect. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the tapered inner diameter may be changed to create the desired locking effect. For example, one or more of the inner diameters 618, 620 may be made larger or smaller to accommodate different sizes of sheaths 106. In another example, the outer diameter of the locking element 600 may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element 600 may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases,

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the sheath 106 and/or the locking element 600 may include visual indicia to guide the user in manipulation of the locking element 600. In some cases, the overall length of the locking element 600 can be increased or decreased, as desired. It is further contemplated that the locking element 600 may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.

FIG. 23 is a perspective view of another illustrative medical device system 700 in a partially unassembled configuration. The medical device system 700 may include a pusher wire 702, an implant (not explicitly shown), such as, but not limited to, an embolic coil, an introducer sheath 706, and a locking element 708. For simplicity, the implant is described as an embolic coil, but other suitable medical devices transported, delivered, used, released, etc. in a similar manner are also contemplated, including but not limited to, vascular occlusion devices coils, stents, embolic filters, replacement heart valves, other occlusion devices, and/or other medical implants, etc. While the medical device system 400 is described with respect to a particular locking element 708, it should be understood that any of the locking elements described herein, including but not limited to the locking elements 108, 200, 300, 408, 508, 600, 800, may be substituted for the locking element 408.

Embolic coils may be typically introduced into a blood vessel by using a microcatheter (not explicitly shown) that extends from a proximal point outside the patient's body to a distal point near the embolization site. An introducer sheath 706 containing the coil can be used to carry and protect the coil prior to insertion into the patient. Further, the introducer sheath 706 may be used to transfer the coil to the microcatheter and/or to assist in deploying the coil at a selected embolization site. The sheath 706 may be configured to protect the implant and maintain the implant in a deliverable orientation, until the implant is deployed. As will be described in more detail herein, the locking element 708 may be configured to limit movement (e.g., axial and rotational) of the pusher wire 702 and implant within the sheath 706 until the user is ready to advance the implant out of the sheath 706.

The sheath 706 may be a tubular member including a proximal end 712, a distal end 714, and an intermediate region 716 positioned therebetween. Some suitable but non-limiting materials for the sheath 706, for example, polymer materials, composite materials, etc., are described below. The sheath 706 may define a lumen 710 extending from the proximal end 712 to the distal end 714. The pusher wire 702 and implant may be slidably disposed within the lumen 710 of the sheath 706 such that the pusher wire 702 and the implant are radially inwards of the sheath 706. The implant may be disposed proximate the distal end 714 of the sheath 706. The pusher wire 702 may be axially slidable between an interlocked position and a released position. The pusher wire 702 may be configured to be releasably attached to the implant. The implant may be configured to expand from a delivery configuration to a deployed configuration. The pusher wire 702 may generally be a solid wire or shaft, but may also be tubular in some embodiments. Some suitable but non-limiting materials for the pusher wire 702, for example metallic materials, polymer materials, composite materials, etc., are described below. As will be described in more detail herein, the pusher wire 702 may be releasably secured to the sheath 706 via the locking element 708 to limit axial and/or rotational movement of the pusher wire 702 within the sheath 706.

The sheath **706** may have a first slot **718a** and a second slot **718b** (collectively, **718**) extending distally from the proximal end **712**. The first and second slots **718a**, **718b** may be positioned across from another or spaced approximately 180° about the circumference of the sheath **706**. While the sheath **706** is described as having two slots, the sheath **706** may include fewer than two or more than two slots, as desired. The slots **718** may extend less than an entire length of the sheath **706**. The slots **718** may remove material from the sheath **706** to create flexible arms or members **720a**, **720b** (collectively, **720**). While the sheath **706** is described as having two arms **720**, it should be understood the number of flexible arms may vary with the number of slots **718** and there can be fewer than two or more than two flexible arms, as desired. Further, the slots **718** may be uniformly or eccentrically distributed about a circumference of the sheath **706**. The length and/or size of the slots **718** (and/or the arms **720**) may vary to produce different degrees of wedging (between the locking element **708** and the pusher wire **702**) and locking capabilities.

The sheath **706** may further include a raised projection **730** extending radially from an outer surface thereof. The raised projection **730** may be positioned adjacent to the proximal end region **729** of the sheath **706**. In some cases, the raised projection **730** may extend distally from the proximal end **712** of the sheath **706**. However, this is not required. The raised projection **730** may be a rectangular prism, such as, but not limited to a ridge or other three-dimensional structure configured to engage a mating groove, notch, or recess within the locking element **708**, as will be described in more detail herein.

It is contemplated that a proximal end region **729** of the sheath **706** may be formed from polypropylene or a material similar thereto. The durometer of the proximal end region **729** may be manipulated to produce a desired locking effect. In some cases, other portions of the sheath **706** may be formed from a same material as the proximal end region **729** while in other cases, other portions of the sheath **706** may be formed from a different material as the proximal end region **729**. For example, the sheath **706** may include a polyimide distal tip (although this is not required).

Referring additionally to FIG. **24**, which illustrates a perspective view of the locking element **708**, the locking element **708** may be a tubular member having a proximal end **722**, a distal end **724**, and an intermediate region **726** positioned therebetween. The locking element **708** may have a generally hourglass shaped outer profile. For example, the outer diameter may generally decrease from the proximal end **722** towards the intermediate region **726** and increase from the intermediate region **726** to the distal end **724**. However, this is not required. The outer profile of the locking element **708** may take any shape desired.

The locking element **708** may define a lumen **728** extending from the proximal end **722** to the distal end **724**. As will be discussed in more detail herein, a portion of the lumen **728** of the locking element **708** may be sized to slide freely over the sheath **706** while another portion of the lumen **728** may be sized to exert a radially inward compressing force on the sheath **706**, as will be described in more detail herein. Thus, in some configurations, at least a portion of the sheath **706** may be radially inward of the locking element **708**.

The intermediate region **726** may include plurality of raised regions or circumferential ribs **732a**, **732b**, **732c** (collectively, **732**). The circumferential ribs **732** may have an outer diameter **734** that is greater than an outer diameter **736** of the intermediate region **726** between the ribs **732**. The circumferential ribs **732** may extend about an entire perim-

eter of the locking element **708** or less than an entire perimeter, as desired. It is contemplated that the circumferential ribs **732** may increase the tactility of the locking element **708** and/or make the locking element **708** easier to grip or handle. The number of circumferential ribs **732** (e.g., fewer than three or greater than three), the size of the circumferential ribs **732** (e.g., increasing or decreasing the diameter **734**), the geometry of the circumferential ribs **732** (extending about less than an entire perimeter, having a different shape, etc.), and/or the spacing of the circumferential ribs **732** can be varied to enhance or diminish the tactility or gripability of the locking element **708**, as desired.

The locking element **708** may further include one or more features to provide a visual cue as to how the locking element **708** should be assembled with the sheath **706**. For example, the feature may be configured to provide an indication to the user how to align the locking element **708** with the sheath **706** such that the locking element **708** may be slid over the sheath **706**. In some cases, the feature may be incorporated into the structure of the locking element **708**. For example, in the embodiment illustrated in FIG. **24**, the locking element **708** includes an axially extending (e.g., parallel to a longitudinal axis of the locking element **708**) raised portion **738**. The raised portion **738** is radially aligned with an internal groove **740** (see, for example, FIGS. **25** and **26**) formed in an inner surface of the locking element **708**. As will be described in more detail herein, the groove **740** may be configured to align with the raised projection **730** of the sheath **706** to allow for selective assembly of the locking element **708** with the sheath **706**. The raised portion **738** is just one example of a feature that may be used to provide visual cues to the operator. Other features may include visual markings on an outer surface of the locking element **708** (e.g., arrows, words, etc.) or other structural features, as desired.

The locking element **708** may be formed in a variety of different manners. For example, the locking element **708** may be injection molded, heat shrunk, 3-D printed, etc. The locking element **708** may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc. It is further contemplated that the locking element **708** may be formed from a material that is more rigid than or stiffer than the outer sheath **706**. For example, the material for the locking element **708** may be selected such that the locking element **708** does not deflect when the tapered portion of the lumen **728** of the locking element **708** is advanced over the sheath **706** but rather forces the sheath **706** to deflect inward. In some embodiments, the locking element **708** may be formed from a bright (or other) color easily noticeable by the user.

FIG. **25** is a cross-sectional view of the locking element **708** taken at line **25-25** of FIG. **24**. FIG. **26** is a distal end view of the locking element **708**. The locking element **708** may have a variable inner cross-sectional dimension. In some cases, the lumen **728** may have a generally tapered configuration. For example, in some embodiments, the lumen **728** of the locking element **708** may have a first cross-sectional dimension **742** adjacent to the proximal end **722** thereof which increases in the distal direction to a second cross-sectional dimension **744** adjacent a distal end **724** thereof which is greater than the first cross-sectional dimension **742**. The cross-sectional dimension may gradually transition in a tapered or sloped manner. In other embodiments, the cross-sectional dimension may vary in an abrupt or step-wise manner. These are just some examples. In some cases, a wall thickness of the locking element **708** may vary over at least a portion of the locking element.

However, this is not required. The dimensions (e.g., inner diameter, outer diameter, length, slope, etc.) of the locking element **708** may be changed to create the desired locking effect (described in more detail herein). The distal inner cross-sectional dimension **744** may be sized such that the locking element **708** can slide freely over the sheath **706**. The proximal, smaller inner cross-sectional dimension **742** may be sized to apply a compressive, or radially inward force on the outer surface of the sheath **706**.

It is contemplated that the configuration of the locking element **708** may be adjusted to create the desired effect. For example, one or more of the inner cross-sectional dimensions **742**, **744** may be made larger or smaller to accommodate different sizes of sheaths **706**. In another example, the outer diameter of the locking element **708** may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element **708** may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath **706** and/or the locking element **708** may include visual indicia to guide the user in manipulation of the locking element **708**. In some cases, the overall length of the locking element **708** can be increased or decreased, as desired. It is further contemplated that the locking element **708** may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.

As described above, the locking element **708** may further include an axially extending (e.g., parallel to a longitudinal axis of the locking element) notch or groove **740**. As shown in FIG. 25, the groove **740** may have a first depth **746** adjacent to the distal end **724** and a second depth **748** adjacent to the proximal end **722**. The second depth **748** is less than the first depth **746** such that a distal portion of the locking element **708** can freely slide over the sheath **706** when the groove **740** is aligned with the raised projection **730** of the sheath **706**. As the locking element **708** is distally advanced, the lumen **728** exerts a radially inward or compressive force on the sheath **706**.

Together, the radial projection **730** of the sheath **706** and the groove **740** of the locking element **708** may create an orientation dependent assembly. For example, the locking element **708** is oriented such that the groove **740** is aligned with the radial projection **730** of the sheath to allow the locking element **708** to distally advance over the sheath **706**. Misalignment of the groove **740** and radial projection **730** may lead to an interference between the locking element **708** and the sheath **706** that would prevent passage of the sheath **706** into the locking element **708** and subsequent locking of the pusher wire **702**. This may prevent undesired re-engagement of the locking element **708** and sheath **706** during use of the system **700** in the event the locking element **708** is not fully removed from the pusher wire **702**. In the absence of orientation dependent features, if the locking element **708** is left hanging on the pusher wire **702**, it may unintentionally re-engage with the sheath **706** and the pusher wire **702** during advancement of the pusher wire **702** through the sheath **706**. It is contemplated that other orientation dependent features may be used on the sheath **706** and/or the locking element **708**, as desired.

FIG. 27 illustrates a perspective view of another illustrative locking element **800** which may include features configured to orientate the locking element **800** relative to a sheath, such as, but not limited to, the sheaths **106**, **406**, **506**, **706**, **706** described herein. The locking element **800** may be a tubular member having a proximal end **802**, a distal end **804**, and an intermediate region **806** positioned therebe-

tween. The locking element **800** may have a generally hourglass shaped outer profile. For example, the outer diameter may generally decrease from the proximal end **802** towards the intermediate region **806** and increase from the intermediate region **806** to the distal end **804**. However, this is not required. The outer profile of the locking element **800** may take any shape desired.

The locking element **800** may define a lumen **808** extending from the proximal end **802** to the distal end **804**. A portion of the lumen **808** of the locking element **800** may be sized to slide freely over a sheath while another portion of the lumen **808** may be sized to exert a radially inward compressing force on the sheath. Thus, in some configurations, at least a portion of the sheath **706** may be radially inward of the locking element **800**.

The intermediate region **806** may include plurality of raised regions or circumferential ribs **810a**, **810b**, **810c** (collectively, **810**). The circumferential ribs **810** may have an outer diameter **812** (see, for example, FIG. 28) that is greater than an outer diameter **814** (see, for example, FIG. 28) of the intermediate region **806** between the ribs **810**. The circumferential ribs **810** may extend about an entire perimeter of the locking element **800** or less than an entire perimeter, as desired. It is contemplated that the circumferential ribs **810** may increase the tactility of the locking element **800** and/or make the locking element **800** easier to grip or handle. The number of circumferential ribs **810** (e.g., fewer than three or greater than three), the size of the circumferential ribs **810** (e.g., increasing or decreasing the diameter **812**), the geometry of the circumferential ribs **810** (extending about less than an entire perimeter, having a different shape, etc.), and/or the spacing of the circumferential ribs **810** can be varied to enhance or diminish the tactility or gripability of the locking element **800**, as desired.

The locking element **800** may be formed in a variety of different manners. For example, the locking element **800** may be injection molded, heat shrunk, 3-D printed, etc. The locking element **800** may be formed from a variety of different materials such as, but not limited to, hard or soft polymers, metals, composites, etc. It is further contemplated that the locking element **800** may be formed from a material that is more rigid than or stiffer than the outer sheath **706**. For example, the material for the locking element **800** may be selected such that the locking element **800** does not deflect when the tapered portion of the lumen **808** of the locking element **800** is advanced over the sheath **706** but rather forces the sheath **706** to deflect inward. In some embodiments, the locking element **800** may be formed from a bright (or other) color easily noticeable by the user.

FIG. 28 is a distal end perspective cross-sectional view of the locking element **800** taken at line 28-28 of FIG. 27. FIG. 29 is a distal end view of the locking element **800**. The locking element **800** may have a variable inner cross-sectional dimension. In some cases, the lumen **808** may have a generally tapered configuration. For example, in some embodiments, the lumen **808** of the locking element **800** may have a first cross-sectional dimension **816** adjacent to the proximal end **802** thereof which increases in the distal direction to a second cross-sectional dimension **818** adjacent to a distal end **804** thereof which is greater than the first cross-sectional dimension **816**. The cross-sectional dimension may gradually transition in a tapered or sloped manner. In other embodiments, the cross-sectional dimension may vary in an abrupt or step-wise manner. These are just some examples. In some cases, a wall thickness of the locking element **800** may vary over at least a portion of the locking element. However, this is not required. The dimensions (e.g.,

inner diameter, outer diameter, length, slope, etc.) of the locking element **800** may be changed to create the desired locking effect (described in more detail herein). The distal inner cross-sectional dimension **818** may be sized such that the locking element **800** can slide freely over the sheath. The proximal, smaller inner cross-sectional dimension **816** may be sized to apply a compressive, or radially inward force on the outer surface of the sheath.

It is contemplated that the configuration of the locking element **800** may be adjusted to create the desired effect. For example, one or more of the inner cross-sectional dimensions **816**, **818** may be made larger or smaller to accommodate different sizes of sheaths. In another example, the outer diameter of the locking element **800** may be increased or decreased to facilitate handling. It is further contemplated that the outer surface of the locking element **800** may include features to improve ergonomic handling, such as, but not limited to, bumps, waves, texturing, or indentations to improve gripability. In some cases, the sheath and/or the locking element **800** may include visual indicia to guide the user in manipulation of the locking element **800**. In some cases, the overall length of the locking element **800** can be increased or decreased, as desired. It is further contemplated that the locking element **800** may be large enough to make it easy to handle but not so large as to be incompatible with other components, such as, but not limited to packaging.

The locking element **800** may further include an axially extending (e.g., parallel to a longitudinal axis of the locking element) protrusion **820** which extends radially inward from an inner surface of the locking element **80**. The protrusion **820** may be a rectangular prism, such as, but not limited to a ridge or other three-dimensional structure configured to engage a mating groove, notch, or recess within the sheath. The protrusion **820** may be sized and shaped such that a distal portion of the locking element **800** can freely slide over the sheath when the protrusion **820** is aligned with a mating slot or recess of the sheath. As the locking element **800** is distally advanced, the lumen **808** narrows so as to exert a radially inward or compressive force on the sheath.

Together, the slot of the sheath and the protrusion **820** of the locking element **800** may create an orientation dependent assembly. For example, the locking element **800** is oriented such that the protrusion **820** is aligned with the slot of the sheath to allow the locking element **800** to distally advance over the sheath. Misalignment of the protrusion **820** and groove may lead to an interference between the locking element **800** and the sheath that would prevent passage of the sheath into the locking element **800** and subsequent locking of the pusher wire. This may prevent undesired re-engagement of the locking element **800** and sheath during use of the system in the event the locking element **800** is not fully removed from the pusher wire. In the absence of orientation dependent features, if the locking element **800** is left hanging on the pusher wire, it may unintentionally re-engage with the sheath and the pusher wire during advancement of the pusher wire through the sheath. It is contemplated that other orientation dependent features may be used on the sheath and/or the locking element **800**, as desired.

While not explicitly shown, the locking element **800** may further include one or more features to provide a visual cue as to how the locking element **800** should be assembled with the sheath. For example, the features may include, but are not limited to, visual markings on an outer surface of the locking element **800** (e.g., arrows, words, etc.) or structural features, as desired.

In some embodiments, the medical device systems **100**, **400**, **500**, **700** the pusher wire **102**, **402**, **502**, **702**, the

implant **104**, the sheath **106**, **406**, **506**, **706**, the locking element **108**, **200**, **300**, **408**, **508**, **600**, **708**, **800**, and/or components thereof, may be made from a metal, metal alloy, polymer (some examples of which are disclosed below), a metal-polymer composite, ceramics, combinations thereof, and the like, or other suitable material. Some examples of suitable metals and metal alloys include stainless steel, such as 444V, 444L, and 314LV stainless steel; mild steel; nickel-titanium alloy such as linear-elastic and/or super-elastic nitinol; other nickel alloys such as nickel-chromium-molybdenum alloys (e.g., UNS: N06625 such as INCONEL® 625, UNS: N06022 such as HASTELLOY® C-22®, UNS: N10276 such as HASTELLOY® C276®, other HASTELLOY® alloys, and the like), nickel-copper alloys (e.g., UNS: N04400 such as MONEL® 400, NICKELVAC® 400, NICORROS® 400, and the like), nickel-cobalt-chromium-molybdenum alloys (e.g., UNS: R44035 such as MP35-N® and the like), nickel-molybdenum alloys (e.g., UNS: N10665 such as HASTELLOY® ALLOY B2®), other nickel-chromium alloys, other nickel-molybdenum alloys, other nickel-cobalt alloys, other nickel-iron alloys, other nickel-copper alloys, other nickel-tungsten or tungsten alloys, and the like; cobalt-chromium alloys; cobalt-chromium-molybdenum alloys (e.g., UNS: R44003 such as ELGILOY®, PHYNOX®, and the like); platinum enriched stainless steel; titanium; platinum; palladium; gold; combinations thereof; and the like; or any other suitable material.

As alluded to herein, within the family of commercially available nickel-titanium or nitinol alloys, is a category designated “linear elastic” or “non-super-elastic” which, although may be similar in chemistry to conventional shape memory and super elastic varieties, may exhibit distinct and useful mechanical properties. Linear elastic and/or non-super-elastic nitinol may be distinguished from super elastic nitinol in that the linear elastic and/or non-super-elastic nitinol does not display a substantial “superelastic plateau” or “flag region” in its stress/strain curve like super elastic nitinol does. Instead, in the linear elastic and/or non-super-elastic nitinol, as recoverable strain increases, the stress continues to increase in a substantially linear, or a somewhat, but not necessarily entirely linear relationship until plastic deformation begins or at least in a relationship that is more linear than the super elastic plateau and/or flag region that may be seen with super elastic nitinol. Thus, for the purposes of this disclosure linear elastic and/or non-super-elastic nitinol may also be termed “substantially” linear elastic and/or non-super-elastic nitinol.

In some cases, linear elastic and/or non-super-elastic nitinol may also be distinguishable from super elastic nitinol in that linear elastic and/or non-super-elastic nitinol may accept up to about 2-5% strain while remaining substantially elastic (e.g., before plastically deforming) whereas super elastic nitinol may accept up to about 8% strain before plastically deforming. Both of these materials can be distinguished from other linear elastic materials such as stainless steel (that can also be distinguished based on its composition), which may accept only about 0.2 to 0.44 percent strain before plastically deforming.

In some embodiments, the linear elastic and/or non-super-elastic nickel-titanium alloy is an alloy that does not show any martensite/austenite phase changes that are detectable by differential scanning calorimetry (DSC) and dynamic metal thermal analysis (DMTA) analysis over a large temperature range. For example, in some embodiments, there may be no martensite/austenite phase changes detectable by DSC and DMTA analysis in the range of about -60 degrees Celsius (° C.) to about 120° C. in the linear elastic and/or

non-super-elastic nickel-titanium alloy. The mechanical bending properties of such material may therefore be generally inert to the effect of temperature over this very broad range of temperature. In some embodiments, the mechanical bending properties of the linear elastic and/or non-super-elastic nickel-titanium alloy at ambient or room temperature are substantially the same as the mechanical properties at body temperature, for example, in that they do not display a super-elastic plateau and/or flag region. For example, across a broad temperature range, the linear elastic and/or non-super-elastic nickel-titanium alloy maintains its linear elastic and/or non-super-elastic characteristics and/or properties.

In some embodiments, the linear elastic and/or non-super-elastic nickel-titanium alloy may be in the range of about 50 to about 60 weight percent nickel, with the remainder being essentially titanium. In some embodiments, the composition is in the range of about 54 to about 57 weight percent nickel. One example of a suitable nickel-titanium alloy is FHP-NT alloy commercially available from Furukawa Techno Material Co. of Kanagawa, Japan. Other suitable materials may include ULTANIUM™ (available from Neo-Metrics) and GUM METAL™ (available from Toyota). In some other embodiments, a superelastic alloy, for example a superelastic nitinol can be used to achieve desired properties. In at least some embodiments, portions or all of the medical device systems 100, 400, 500, 700, the pusher wire 102, 402, 502, 702, the implant 104, the sheath 106, 406, 506, 706, the locking element 108, 200, 300, 408, 508, 600, 708, 800, and/or components thereof, may also be doped with, made of, or otherwise include a radiopaque material. Radiopaque materials are understood to be materials capable of producing a relatively bright image on a fluoroscopy screen or another imaging technique during a medical procedure. This relatively bright image aids a user in determining the location of the medical device systems 100, 400, 500, 700, the pusher wire 102, 402, 502, 702, the implant 104, the sheath 106, 406, 506, 706, the locking element 108, 200, 300, 408, 508, 600, 708, 800, and/or components thereof, etc. Some examples of radiopaque materials can include, but are not limited to, gold, platinum, palladium, tantalum, tungsten alloy, polymer material loaded with a radiopaque filler, and the like. Additionally, other radiopaque marker bands and/or coils may also be incorporated into the design of the medical device systems 100, 400, 500, 700, the pusher wire 102, 402, 502, 702, the implant 104, the sheath 106, 406, 506, 706, the locking element 108, 200, 300, 408, 508, 600, 708, 800, and/or components thereof, etc. to achieve the same result.

In some embodiments, a degree of Magnetic Resonance Imaging (MRI) compatibility is imparted into the medical device systems 100, 400, 500, 700, the pusher wire 102, 402, 502, 702, the implant 104, the sheath 106, 406, 506, 706, the locking element 108, 200, 300, 408, 508, 600, 708, 800, and/or components thereof, etc. For example, the medical device system 100, the pusher wire 102, the implant 104, the sheath 106, the locking element 108, 200, etc., and/or components or portions thereof, may be made of a material that does not substantially distort the image and create substantial artifacts (e.g., gaps in the image). Certain ferromagnetic materials, for example, may not be suitable because they may create artifacts in an MRI image. The medical device system 100, the pusher wire 102, the implant 104, the sheath 106, the locking element 108, 200, etc., or portions thereof, may also be made from a material that the MRI machine can image. Some materials that exhibit these characteristics include, for example, tungsten, cobalt-chromium-molybdenum alloys (e.g., UNS: R44003 such as ELGILOY®, PHYNOX®, and the like), nickel-cobalt-chro-

mium-molybdenum alloys (e.g., UNS: R44035 such as MP35-N® and the like), nitinol, and the like, and others.

In some embodiments, the medical device systems 100, 400, 500, 700, the pusher wire 102, 402, 502, 702, the implant 104, the sheath 106, 406, 506, 706, the locking element 108, 200, 300, 408, 508, 600, 708, 800, and/or components thereof, may be made from or include a polymer or other suitable material. Some examples of suitable polymers may include polytetrafluoroethylene (PTFE), ethylene tetrafluoroethylene (ETFE), fluorinated ethylene propylene (FEP), polyoxymethylene (POM, for example, DELRIN® available from DuPont), polyether block ester, polyurethane (for example, Polyurethane 85A), polypropylene (PP), polyvinylchloride (PVC), polyether-ester (for example, ARNITEL® available from DSM Engineering Plastics), ether or ester based copolymers (for example, butylene/poly(alkylene ether) phthalate and/or other polyester elastomers such as HYTREL® available from DuPont), polyamide (for example, DURETHAN® available from Bayer or CRISTAMID® available from Elf Atochem), elastomeric polyamides, block polyamide/ethers, polyether block amide (PEBA, for example available under the trade name PEBAX®), ethylene vinyl acetate copolymers (EVA), silicones, polyethylene (PE), Marlex® high-density polyethylene, Marlex® low-density polyethylene, linear low density polyethylene (for example REXELL®), polyester, polybutylene terephthalate (PBT), polyethylene terephthalate (PET), polytrimethylene terephthalate, polyethylene naphthalate (PEN), polyetheretherketone (PEEK), polyimide (PI), polyetherimide (PEI), polyphenylene sulfide (PPS), polyphenylene oxide (PPO), poly paraphenylene terephthalamide (for example, KEVLAR®), polysulfone, nylon, nylon-12 (such as GRILAMID® available from EMS American Grilon), perfluoro(propyl vinyl ether) (PFA), ethylene vinyl alcohol, polyolefin, polystyrene, epoxy, polyvinylidene chloride (PVdC), poly(styrene-b-isobutylene-b-styrene) (for example, SIBS and/or SIBS 50A), polycarbonates, ionomers, biocompatible polymers, other suitable materials, or mixtures, combinations, copolymers thereof, polymer/metal composites, and the like. In some embodiments the sheath can be blended with a liquid crystal polymer (LCP). For example, the mixture can contain up to about 6 percent LCP.

In some embodiments, the medical device systems 100, 400, 500, 700, the pusher wire 102, 402, 502, 702, the implant 104, the sheath 106, 406, 506, 706, the locking element 108, 200, 300, 408, 508, 600, 708, 800, and/or components thereof, etc. disclosed herein may include a fabric material disposed over or within the structure. The fabric material may be composed of a biocompatible material, such a polymeric material or biomaterial, adapted to promote tissue ingrowth. In some embodiments, the fabric material may include a bioabsorbable material. Some examples of suitable fabric materials include, but are not limited to, polyethylene glycol (PEG), nylon, polytetrafluoroethylene (PTFE, ePTFE), a polyolefinic material such as a polyethylene, a polypropylene, polyester, polyurethane, and/or blends or combinations thereof.

In some embodiments, the medical device systems 100, 400, 500, 700, the pusher wire 102, 402, 502, 702, the implant 104, the sheath 106, 406, 506, 706, the locking element 108, 200, 300, 408, 508, 600, 708, 800, and/or components thereof, etc. may include and/or be formed from a textile material. Some examples of suitable textile materials may include synthetic yarns that may be flat, shaped, twisted, textured, pre-shrunk or un-shrunk. Synthetic biocompatible yarns suitable for use in the present invention

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include, but are not limited to, polyesters, including polyethylene terephthalate (PET) polyesters, polypropylenes, polyethylenes, polyurethanes, polyolefins, polyvinyl s, polymethylacetates, polyamides, naphthalene dicarboxylene derivatives, natural silk, and polytetrafluoroethylenes. Moreover, at least one of the synthetic yarns may be a metallic yarn or a glass or ceramic yarn or fiber. Useful metallic yarns include those yarns made from or containing stainless steel, platinum, gold, titanium, tantalum or a Ni—Co—Cr-based alloy. The yarns may further include carbon, glass or ceramic fibers. Desirably, the yarns are made from thermoplastic materials including, but not limited to, polyesters, polypropylenes, polyethylenes, polyurethanes, polynaphthalenes, polytetrafluoroethylenes, and the like. The yarns may be of the multifilament, monofilament, or spun-types. The type and denier of the yarn chosen may be selected in a manner which forms a biocompatible and implantable prosthesis and, more particularly, a vascular structure having desirable properties.

In some embodiments, the medical device systems **100**, **400**, **500**, **700**, the pusher wire **102**, **402**, **502**, **702**, the implant **104**, the sheath **106**, **406**, **506**, **706**, the locking element **108**, **200**, **300**, **408**, **508**, **600**, **708**, **800**, and/or components thereof, etc. may include and/or be treated with a suitable therapeutic agent. Some examples of suitable therapeutic agents may include anti-thrombogenic agents (such as heparin, heparin derivatives, urokinase, and PPACK (dextrophenylalanine proline arginine chloromethylketone)); anti-proliferative agents (such as enoxaparin, angiopentin, monoclonal antibodies capable of blocking smooth muscle cell proliferation, hirudin, and acetylsalicylic acid); anti-inflammatory agents (such as dexamethasone, prednisolone, corticosterone, budesonide, estrogen, sulfasalazine, and mesalamine); antineoplastic/antiproliferative/anti-mitotic agents (such as paclitaxel, 5-fluorouracil, cisplatin, vinblastine, vincristine, epothilones, endostatin, angiostatin and thymidine kinase inhibitors); anesthetic agents (such as lidocaine, bupivacaine, and ropivacaine); anti-coagulants (such as D-Phe-Pro-Arg chloromethyl ketone, an RGD peptide-containing compound, heparin, anti-thrombin compounds, platelet receptor antagonists, anti-thrombin antibodies, anti-platelet receptor antibodies, aspirin, prostaglandin inhibitors, platelet inhibitors, and tick antiplatelet peptides); vascular cell growth promoters (such as growth factor inhibitors, growth factor receptor antagonists, transcriptional activators, and translational promoters); vascular cell growth inhibitors (such as growth factor inhibitors, growth factor receptor antagonists, transcriptional repressors, translational repressors, replication inhibitors, inhibitory antibodies, antibodies directed against growth factors, bifunctional molecules consisting of a growth factor and a cytotoxin, bifunctional molecules consisting of an antibody and a cytotoxin); cholesterol-lowering agents; vasodilating agents; and agents which interfere with endogenous vasoactive mechanisms.

It should be understood that this disclosure is, in many respects, only illustrative. Changes may be made in details, particularly in matters of shape, size, and arrangement of steps without exceeding the scope of the invention. This may include, to the extent that it is appropriate, the use of any of the features of one example embodiment being used in other embodiments. The invention's scope is, of course, defined in the language in which the appended claims are expressed.

What is claimed is:

1. A medical device system, comprising:
  - a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the

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distal end, and a lumen extending from the proximal end to the distal end, the sheath further comprising a longitudinally extending slot extending distally from the proximal end of the sheath, the longitudinally extending slot terminating proximally of the intermediate region;

- a pusher wire slidably disposed within the lumen of the sheath, wherein a distal end of the pusher wire is configured to releasably engage an embolic coil positioned within the distal region of the lumen of the sheath; and
- a locking element configured to non-threadably engage the sheath, the locking element having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end of the locking element, and a lumen extending from the proximal end to the distal end of the locking element, the lumen of the locking element having a generally hourglass shape;
  - wherein a distal end region of the locking element is configured to freely slide over the proximal end of the sheath;
  - wherein when the intermediate region of the locking element is disposed over the proximal end of the sheath, the locking element is configured to depress the sheath radially inwards onto a proximal end of the pusher wire.

2. The medical device system of claim 1, wherein the longitudinally extending slot of the sheath terminates at a circumferentially extending slit.

3. The medical device system of claim 1, wherein the sheath further comprises a first flap and a second flap adjacent the longitudinally extending slot.

4. The medical device system of claim 1, wherein the locking element has a first inner diameter adjacent the distal end of the locking element and a second inner diameter adjacent to the intermediate region of the locking element, the second inner diameter smaller than the first inner diameter.

5. The medical device system of claim 4, wherein the second inner diameter is generally constant and extends between a flared distal end region and a flared proximal end region.

6. The medical device system of claim 1, wherein the locking element is symmetrical about a plane oriented perpendicular to a central longitudinal axis of the locking element.

7. The medical device system of claim 1, wherein the locking element has a generally uniform wall thickness from the proximal end to the distal end of the locking element.

8. A medical device system, comprising:

a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the sheath further comprising a pair of longitudinally extending slots each extending distally from the proximal end of the sheath and forming a pair of flexible arms therebetween;

a pusher wire slidably disposed within the lumen of the sheath, wherein a distal end of the pusher wire is configured to releasably engage an implant positioned within the lumen of the sheath; and a locking element having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end of the locking element, and a lumen extending from the proximal end to the distal end of the

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locking element, the lumen of the locking element having a generally hourglass shape;  
 wherein a distal end region of the locking element is configured to freely slide over the proximal end of the sheath;

wherein when the intermediate region of the locking element is disposed over the proximal end of the sheath, the locking element is configured to depress the pair of flexible arms radially inwards against a proximal end of the pusher wire;

wherein the locking element is symmetrical about a plane oriented perpendicular to a central longitudinal axis of the locking element.

9. The medical device system of claim 8, wherein the locking element has a first inner diameter adjacent the distal end of the locking element and a second inner diameter adjacent to the intermediate region of the locking element, the second inner diameter smaller than the first inner diameter.

10. The medical device system of claim 9, wherein the second inner diameter is generally constant and extends between a flared distal end region and a flared proximal end region.

11. The medical device system of claim 8, wherein the locking element has a generally uniform wall thickness from the proximal end to the distal end of the locking element.

12. A medical device system, comprising:

a sheath having a proximal end, a distal end, an intermediate region disposed between the proximal end and the distal end, and a lumen extending from the proximal end to the distal end, the sheath further comprising a longitudinally extending slot extending distally from the proximal end of the sheath;

a pusher wire slidably disposed within the lumen of the sheath, wherein a distal end of the pusher wire is configured to releasably engage an embolic coil positioned within the lumen of the sheath; and

a locking element having a proximal end, a distal end, an intermediate region disposed between the proximal end

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and the distal end of the locking element, and a lumen extending from the proximal end to the distal end of the locking element, the lumen of the locking element having a generally hourglass shape;

wherein a distal end region of the locking element is configured to freely slide over the proximal end of the sheath; and

wherein when the intermediate region of the locking element is disposed over the proximal end of the sheath, the locking element is configured to depress the sheath radially inwards against a proximal end of the pusher wire;

wherein the locking element has a generally uniform wall thickness from the proximal end to the distal end of the locking element.

13. The medical device system of claim 12, wherein the longitudinally extending slot of the sheath terminates at a circumferentially extending slit.

14. The medical device system of claim 12, wherein the sheath further comprises a first flap and a second flap adjacent the longitudinally extending slot.

15. The medical device system of claim 12, wherein the locking element has a first inner diameter adjacent the distal end of the locking element and a second inner diameter adjacent to the intermediate region of the locking element, the second inner diameter smaller than the first inner diameter.

16. The medical device system of claim 15, wherein the second inner diameter is generally constant and extends between a flared distal end region and a flared proximal end region.

17. The medical device system of claim 12, wherein the locking element is symmetrical about a plane oriented perpendicular to a central longitudinal axis of the locking element.

18. The medical device system of claim 12, wherein the locking element is configured to non-threadably engage the sheath.

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